

Unit 14

HYDROCARBONS

132 items · 333 marks

Syllabus sections in this unit

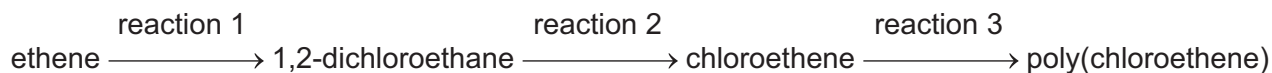
14.1 Alkanes

14.2 Alkenes

Unit 14: Hydrocarbons**1. M/J 26/P21/Q5(b)**

Context

- 5** Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

**Figure 5.1**

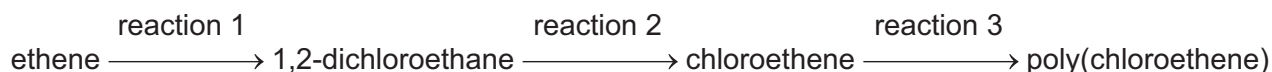
- (b)** In reaction 1, ethene reacts with $\text{Cl}_2(\text{g})$ at room temperature.
Name the mechanism for reaction 1.

[1]

2. M/J 26/P21/Q5(d)

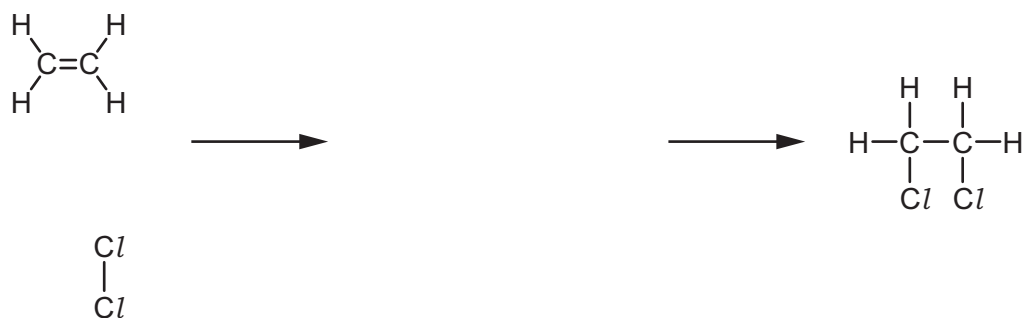
Context

- 5** Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

**Figure 5.1**

- (d)** Complete Figure 5.2 to show the mechanism for the reaction of ethene with Cl_2 in reaction 1.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Figure 5.2**

[4]

3. M/J 26/P22/Q5(b)(iii)-(iv)

Context

- 5** Hexane, $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_5\text{Cl}$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_3$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_2\text{Cl}$	1,2-dichlorohexane
$\text{CH}_3(\text{CH}_2)_{10}\text{CH}_3$	dodecane
$\text{CH}_3(\text{CH}_2)_3\text{CH}(\text{CH}_3)\text{CH}(\text{CH}_3)(\text{CH}_2)_3\text{CH}_3$	5,6-dimethyldecane

Context: part (b) (introduction)

- (b)** The mechanism for the reaction of hexane with $\text{Cl}_2(\text{g})$ involves three stages.

stage 1 initiation

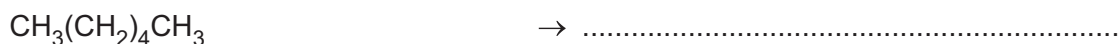
stage 2 propagation

stage 3 termination

In stage 1 and 2, free radicals are produced.

- (iii)** Suggest a reaction sequence of **two** propagation reactions that result in $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$ forming $\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_3$.

propagation reaction 1



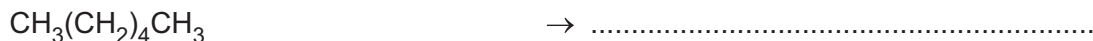
propagation reaction 2



[2]

- (iv)** Suggest a reaction sequence of **one** propagation reaction followed by a termination reaction that results in $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$ forming $\text{CH}_3(\text{CH}_2)_{10}\text{CH}_3$.

propagation reaction



termination reaction



[2]

4. M/J 26/P22/Q5(c)

Context

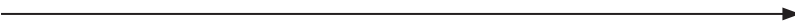
- 5** Hexane, $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_5\text{Cl}$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_3$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_2\text{Cl}$	1,2-dichlorohexane
$\text{CH}_3(\text{CH}_2)_{10}\text{CH}_3$	dodecane
$\text{CH}_3(\text{CH}_2)_3\text{CH}(\text{CH}_3)\text{CH}(\text{CH}_3)(\text{CH}_2)_3\text{CH}_3$	5,6-dimethyldecane

- (c) The proportion of each product varies depending on the type of free radical intermediate that is produced.

Table 5.2

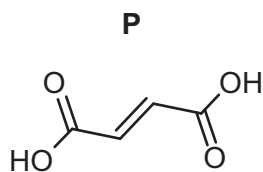
primary free radical	secondary free radical	tertiary free radical
$\bullet\text{C}(\text{CH}_3)\text{H}_2$	$\bullet\text{C}(\text{CH}_3)_2\text{H}$	$\bullet\text{C}(\text{CH}_3)_3$
 increased stability of radical		

Use Table 5.2 to suggest which factor determines the increased stability of free radicals from primary to secondary to tertiary.

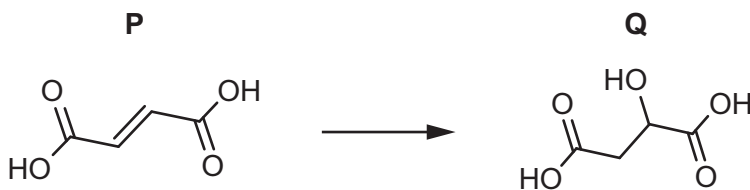
[1]

5. M/J 26/P22/Q6(b)(i)

Context: part (a) (introduction)

6 (a) Figure 6.1 shows the structure of compound **P**.**Figure 6.1**

Context: part (b) (introduction)

(b) **P** reacts to produce compound **Q**.**Figure 6.2****(i)** Identify the reagent and conditions required to convert **P** to **Q**.

reagent

conditions

[2]

6. M/J 26/P23/Q5(a)(ii)

Context

5 Propane, $\text{CH}_3\text{CH}_2\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.**Table 5.1**

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_2\text{Cl}$	1-chloropropane
$\text{CH}_3\text{CHClCH}_3$	2-chloropropane
$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2,3-dimethylbutane
	hexane
	2-methylpentane

(ii) Name the mechanism for the reaction of propane and $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light.

[1]

7. M/J 26/P23/Q5(b)(ii)-(iii)

Context

- 5** Propane, $\text{CH}_3\text{CH}_2\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_2\text{Cl}$	1-chloropropane
$\text{CH}_3\text{CHClCH}_3$	2-chloropropane
$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2,3-dimethylbutane
	hexane
	2-methylpentane

Context: part (b) (introduction)

- (b)** The mechanism for this reaction involves three stages.

stage 1 initiation

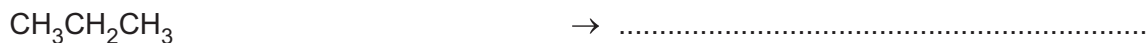
stage 2 propagation

stage 3 termination

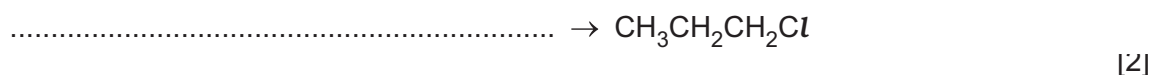
During the initiation reaction, a chlorine free radical is produced.

- (ii)** Suggest a reaction sequence of **two** propagation reactions that result in $\text{CH}_3\text{CH}_2\text{CH}_3$ forming $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$.

propagation reaction 1

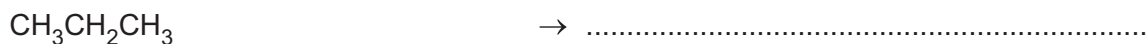


propagation reaction 2

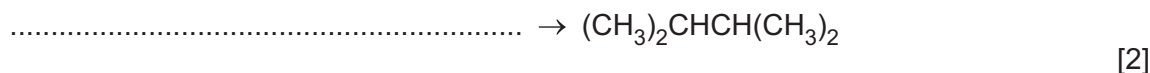


- (iii)** Suggest a reaction sequence of **one** propagation reaction followed by a termination reaction that results in $\text{CH}_3\text{CH}_2\text{CH}_3$ forming $(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$.

propagation reaction 3



termination reaction

**8. M/J 26/P24/Q5(a)**

Context

- 5** Long-chain alkanes from crude oil can be cracked to make alkenes.

- (a)** State the conditions used for the cracking of alkanes.

[1]

9. M/J 26/P24/Q5(b)(iii)

Context

5 Long-chain alkanes from crude oil can be cracked to make alkenes.

Context: part (b) (introduction)

(b) Propene is an alkene that is made by cracking. Alkenes are used in addition polymerisation.

(iii) Aqueous bromine is used to test for the presence of the C=C bond in propene.

State what is observed when aqueous bromine is mixed with propene.

[1]

10. M/J 26/P24/Q5(d)

Context

5 Long-chain alkanes from crude oil can be cracked to make alkenes.

(d) But-2-ene reacts with HBr(g). The HBr acts as an electrophile in the reaction.

(i) State how HBr acts as an electrophile in terms of electron transfer.

[1]

(ii) Complete Figure 5.1 to show the mechanism for the reaction of but-2-ene with HBr.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Figure 5.1

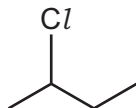
[4]

11. MAR 26/P22/Q3(c)

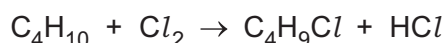
Context

- 3** Secondary halogenoalkanes are important building blocks in organic chemistry.

2-chlorobutane is an example of a secondary halogenoalkane.

2-chlorobutane**Figure 3.1**

- (c) Butane, C_4H_{10} , reacts with Cl_2 in the presence of ultraviolet light to form 2-chlorobutane, which has molecular formula C_4H_9Cl .



- (i) Name the mechanism of this reaction.

[1]

- (ii) Complete the reaction scheme given to show the formation of 2-chlorobutane.

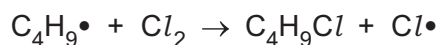
You may use molecular formulas in your answer.

initiation



propagation

propagation



termination

[2]

12. O/N 25/P21/Q3(c)(ii)-(iii)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (c) (introduction)

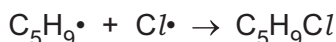
- (c) Cyclopentane reacts with Cl_2 in the presence of ultraviolet light to form C_5H_9Cl .

- (ii) Complete the equations to show the two propagation steps that follow the initiation step.



[2]

- (iii) The final step is shown.



Give the name for this step in the reaction.

[1]

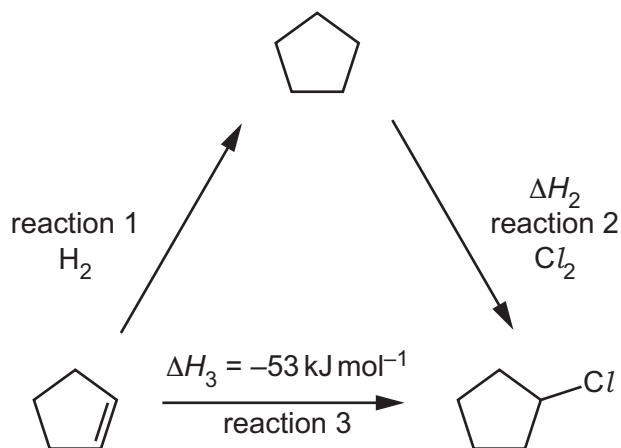
13. O/N 25/P21/Q3(d)(i)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (d) (introduction)

- (d)** Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and C_5H_9Cl .

**Fig. 3.2**

- (i)** Identify a suitable reagent for reaction 3.

[1]

14. O/N 25/P21/Q3(e)(i)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (e) (introduction)

- (e)** Cyclopentene, C_5H_8 , reacts with hot concentrated acidified $KMnO_4$ to form compound **W**, $C_5H_8O_4$.

- (i)** Draw the structure of **W**.

[1]

15. O/N 25/P22/Q3(b)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

- (b)** Give two reasons to explain the general unreactivity of alkanes.

1

2

[2]

16. O/N 25/P22/Q3(c)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

(c) Alkanes with low relative molecular mass, M_r , are more useful than those found in heavier crude oil fractions.

Name the process that is used to obtain alkanes with low M_r from heavier crude oil fractions.

[1]

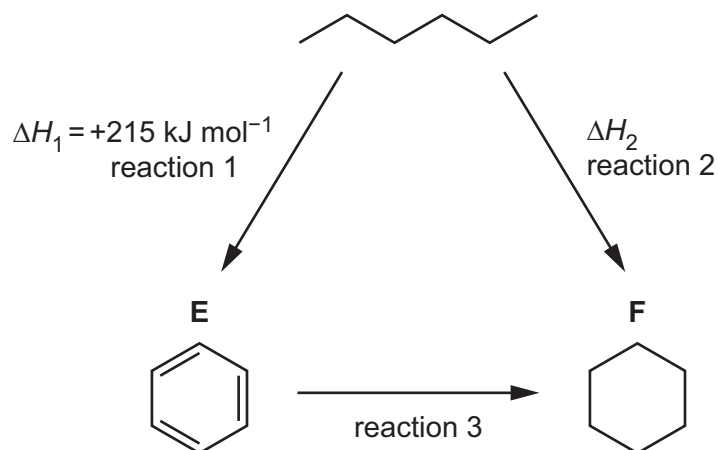
17. O/N 25/P22/Q3(e)(i)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

Context: part (e) (introduction)

(e) Hexane can be converted into compounds **E** and **F** at high temperature and pressure. Fig. 3.2 shows the reaction scheme involving hexane, **E** and **F**.

**Fig. 3.2**

- (i) Identify a suitable reagent for reaction 3.

[1]

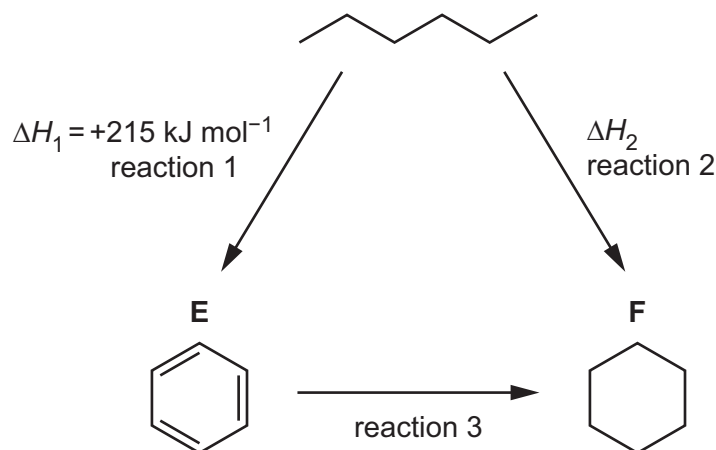
18. O/N 25/P22/Q3(e)(iii)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

Context: part (e) (introduction)

- (e)** Hexane can be converted into compounds **E** and **F** at high temperature and pressure. Fig. 3.2 shows the reaction scheme involving hexane, **E** and **F**.

**Fig. 3.2**

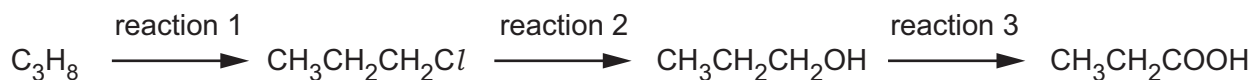
- (iii)** Write an equation for the complete combustion of hexane.

[1]

19. O/N 25/P22/Q4(a)(i)-(ii)

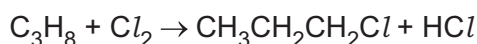
Context

- 4** Fig. 4.1 shows how propane, C_3H_8 , can be converted to propanoic acid, CH_3CH_2COOH .

**Fig. 4.1**

Context: part (a) (introduction)

- (a)** Reaction 1 in Fig. 4.1 takes place in the presence of sunlight.



The reaction takes place via initiation, propagation and termination steps.

- (i)** Name the mechanism shown by reaction 1.

[1]

- (ii)** Complete the mechanism for reaction 1.

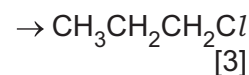
Construct equations to describe the steps of the mechanism.



propagation 1

propagation 2

termination



[3]

20. O/N 25/P23/Q3(c)(ii)-(iii)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (c) (introduction)

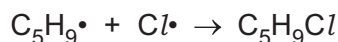
(c) Cyclopentane reacts with Cl_2 in the presence of ultraviolet light to form $\text{C}_5\text{H}_9\text{Cl}$.

- (ii) Complete the equations to show the two propagation steps that follow the initiation step.



[2]

- (iii) The final step is shown.



Give the name for this step in the reaction.

[1]

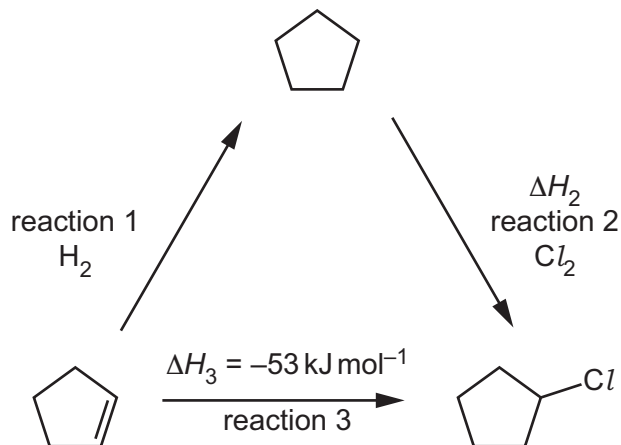
21. O/N 25/P23/Q3(d)(i)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (d) (introduction)

(d) Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and $\text{C}_5\text{H}_9\text{Cl}$.

**Fig. 3.2**

- (i) Identify a suitable reagent for reaction 3.

[1]

22. O/N 25/P23/Q3(e)(i)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (e) (introduction)

- (e)** Cyclopentene, C_5H_8 , reacts with hot concentrated acidified $KMnO_4$ to form compound **W**, $C_5H_8O_4$.

- (i)** Draw the structure of **W**.

[1]

23. O/N 25/P24/Q4(a)

also Unit 13 (4 marks), Unit 22 (2 marks); this unit 5 marks

Context

- 4** Compounds **P** and **Q** are structural isomers.

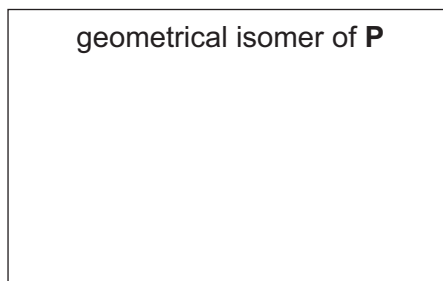
**Fig. 4.1**

- (a) (i)** Define structural isomerism.

[2]

- (ii)** **P** shows geometrical isomerism.

Draw the geometrical isomer of **P**. Explain why the two isomers are **not** identical.



explanation

[2]

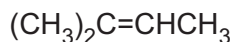
- (iii)** Both **P** and **Q** react with aqueous bromine.

Name the mechanism of this reaction.

[1]

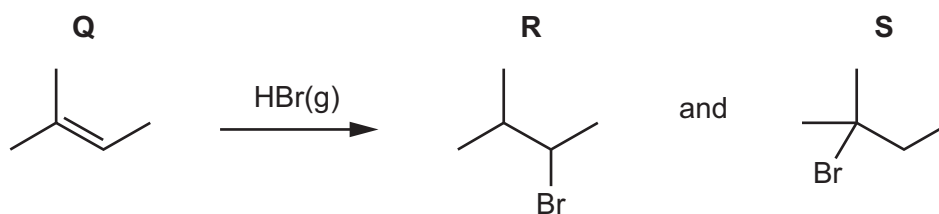
- (iv)** **Q** is oxidised by hot concentrated acidified $KMnO_4(aq)$, forming two different organic products.

Construct an equation for this reaction. Use [O] to represent an atom of oxygen from the oxidising agent.



[2]

- (v)** **Q** reacts with $HBr(g)$ to produce two structural isomers, **R** and **S**, as shown in Fig. 4.2.

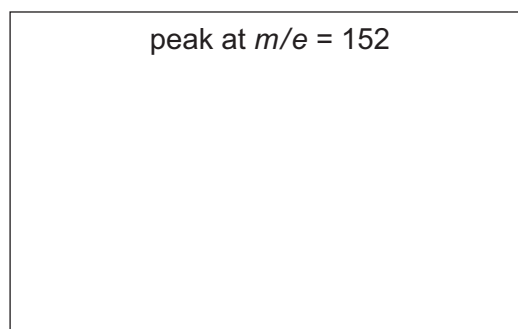
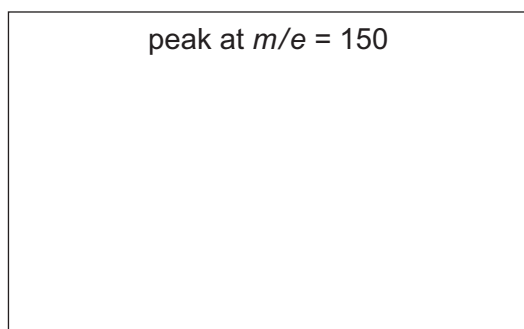
**Fig. 4.2**

State and explain why isomer **S** is the major product of the reaction.

[2]

(vi) The mass spectrum of **R** shows peaks at $m/e = 150$ and $m/e = 152$.

Suggest structures for the ions responsible for these peaks.



[2]

24. M/J 25/P21/Q4(b)(iv)

Context

4 The structure of vitamin C is shown in Fig. 4.1.

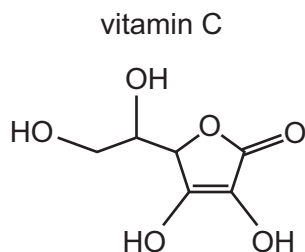


Fig. 4.1

Context: part (b) (introduction)

(b) The concentration of vitamin C is found by titration with $\text{I}_2(\text{aq})$.

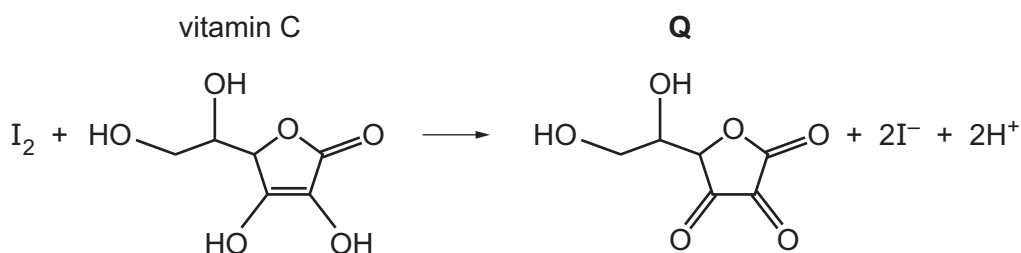


Fig. 4.2

A vitamin C tablet is dissolved in water to produce 200.0 cm³ of vitamin C solution. 5.00 cm³ of this vitamin C solution is added to a flask with approximately 150 cm³ of water and an indicator. Exactly 28.40 cm³ of 5.00 × 10⁻⁴ mol dm⁻³ I₂(aq) reacts with the sample of vitamin C solution in the flask.

[M_r : vitamin C, 176]

(iv) Suggest **two** reasons why hot concentrated acidified potassium manganate(VII) is **not** a suitable reagent for producing **Q** from vitamin C.

1

2

[2]

25. M/J 25/P22/Q3(a)(ii)-(iii)

Context: part (a) (introduction)

3 (a) C_3H_6 reacts with HBr(g) in an addition reaction.

- (ii) Complete Fig. 3.1 to show the mechanism for the addition reaction between C_3H_6 and HBr to produce 2-bromopropane. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Fig. 3.1**

- (iii) Explain why the major product of this reaction is 2-bromopropane rather than 1-bromopropane.

26. M/J 25/P22/Q3(e)

- (e) Alkene **Z** contains two $\text{C}=\text{C}$ bonds. **Z** reacts with an excess of hot concentrated acidified KMnO_4 to produce **only** CH_3COCH_3 , $\text{HOOCCH}_2\text{COOH}$, CO_2 and H_2O .

Suggest the structure of **Z**.

27. M/J 25/P23/Q3(a)(i)

Context: part (a) (introduction)

3 (a) Different hydrocarbon mixtures produced from fractional distillation of crude oil have different uses.

- (i) State the compound that is heated with long-chain hydrocarbons to produce more useful smaller alkanes and alkenes.

28. M/J 25/P23/Q3(b)(i)

Context: part (b) (introduction)

- (b) C_4H_8 reacts with an excess of H_2 to produce C_4H_{10} .



- (i) Name a catalyst for reaction 1.

29. M/J 25/P23/Q6(b)

- (b) Complete Fig. 6.1 to show the mechanism for the reaction between HOCl and $\text{CH}_2=\text{CH}_2$ to produce $\text{HOCH}_2\text{CH}_2\text{Cl}$.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 6.1

[4]

30. M/J 25/P24/Q5(d)

Context

- 5** The skeletal formula of **W** is shown in Fig. 5.1.

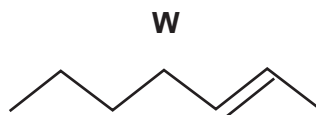


Fig. 5.1

- (d) **W** reacts with reagent **X** to produce ethanoic acid and pentanoic acid.

- (i) State the role of **X** in this reaction.

[1]

- (ii) Identify **X** and state the conditions used for this reaction.

[1]

31. MAR 25/P22/Q3(d)

also Unit 3 (1 mark), Unit 13 (1 mark); this unit 5 marks

Context

- 3** The halogens chlorine, bromine and iodine show trends in chemical and physical properties down the group.

Table 3.1 shows some properties of chlorine, bromine and iodine.

Table 3.1

property	chlorine	bromine	iodine
colour and state at room temperature	green gas		
bond energy / kJ mol^{-1}	242	193	151
electronegativity	3.0	2.8	2.5
formula of sodium halide	NaCl	NaBr	NaI

- (d)** Iodine monobromide, IBr , is a dark red solid that melts near room temperature.

IBr reacts with propene. The mechanism for this reaction is the same as the mechanism for that of HBr with propene.

- (i)** Identify **all** the intermolecular forces that exist between molecules of IBr .

[1]

- (ii)** Name the mechanism involved in the reaction of IBr with propene.

[1]

- (iii)** The reaction of IBr with propene forms two structural isomers.

Draw the **two** structural isomers shown by these molecules.



[2]

- (iv)** Identify the type of structural isomerism shown by the molecules in **(d)(iii)**.

[1]

- (v)** Explain why the two structural isomers do **not** form in equal amounts.

[2]

32. O/N 24/P21/Q2(b)(ii)-(iii)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (b) (introduction)

(b) The halogenoalkane $\text{CH}_3\text{CH}_2\text{Cl}$ forms when chlorine reacts with C_2H_6 via a free-radical substitution mechanism.**(ii)** State the essential condition for chlorine to react with C_2H_6 at room temperature.

[1]

(iii) Write **two** equations to show the propagation steps in this reaction.

1

2

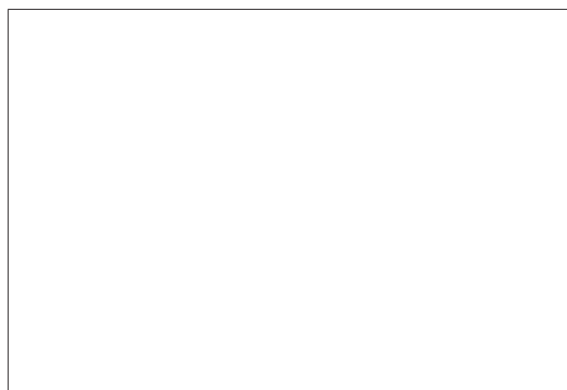
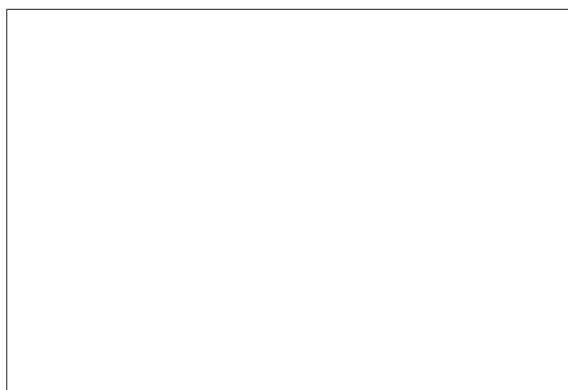
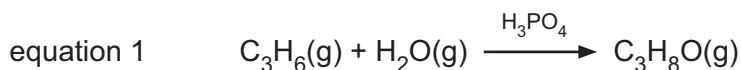
[2]

33. O/N 24/P22/Q2(b)(iii)

Context

2 Oxygen is a Group 16 element.

Context: part (b) (introduction)

(b) H_2O reacts with both inorganic and organic compounds.**(iii)** Draw the structures of the two alcohols formed in the reaction shown in equation 1.

[2]

34. O/N 24/P23/Q2(b)(ii)-(iii)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (b) (introduction)

(b) The halogenoalkane $\text{CH}_3\text{CH}_2\text{Cl}$ forms when chlorine reacts with C_2H_6 via a free-radical substitution mechanism.**(ii)** State the essential condition for chlorine to react with C_2H_6 at room temperature.

[1]

(iii) Write **two** equations to show the propagation steps in this reaction.

1

2

[2]

35. M/J 24/P21/Q5(c)(ii)

Context

5 Hydrocarbon molecules contain covalent bonds.

Context: part (c) (introduction)

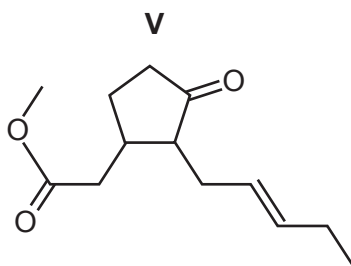
(c) In electrophilic reactions involving alkenes the π bond of $C=C$ is broken.**(ii)** Complete Fig. 5.1 to show the mechanism for the electrophilic addition of hydrogen bromide to 2-methylpropene to produce the major organic product.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

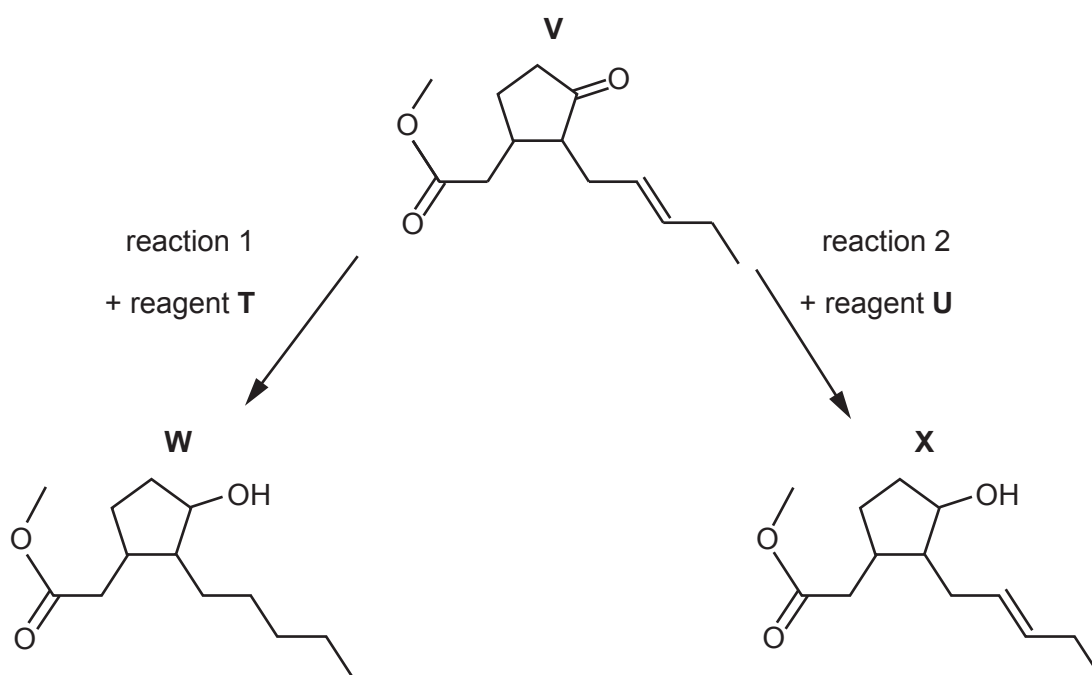
**Fig. 5.1****[4]**

36. M/J 24/P21/Q6(b)(ii)

Context: part (a) (introduction)

6 (a) V shows stereoisomerism.**Fig. 6.1**

Context: part (b) (introduction)

(b) Fig. 6.2 shows two reactions involving **V**.**Fig. 6.2****(ii)** Suggest the identity of reagent **U** in reaction 2.

37. M/J 24/P23/Q5(a)

5 (a) **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

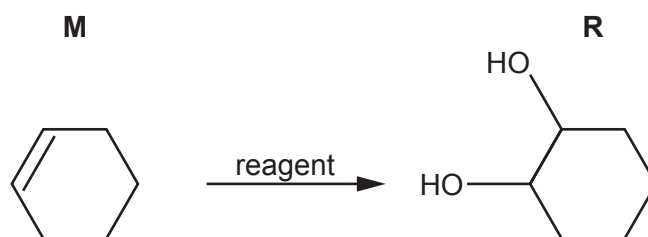


Fig. 5.1

Identify the reagent and conditions for this reaction.

[1]

38. M/J 24/P23/Q5(b)(ii)

Context: part (a)

5 (a) **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

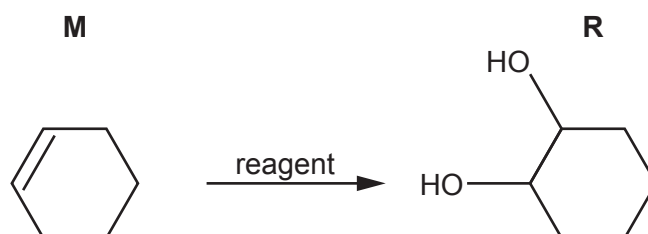


Fig. 5.1

Identify the reagent and conditions for this reaction.

[1]

Context: part (b) (introduction)

(b) **R** is also made from **M** by two steps, as shown in Fig. 5.2.

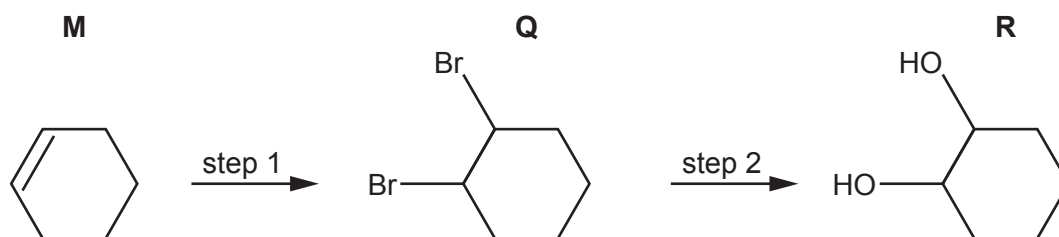


Fig. 5.2

(ii) Name the mechanism for step 1 in Fig. 5.2.

[1]

39. MAR 24/P22/Q1(d)(ii)

Context

- 1** Bismuth is an element in Group 15 of the Periodic Table.

Context: part (d) (introduction)

- (d)** NaBiO_3 is an oxidising agent with similar properties to KMnO_4 .

Fig. 1.1 shows an example of the use of NaBiO_3 as an oxidising agent.

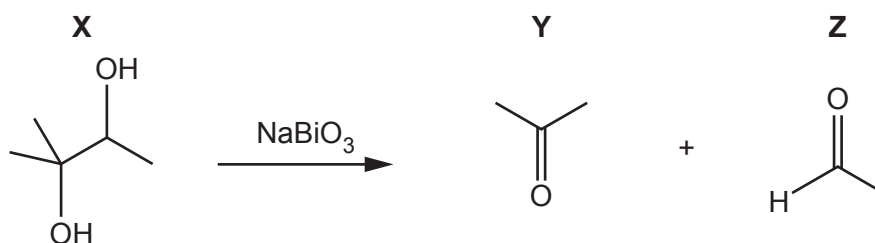


Fig. 1.1

- (ii)** Compound **X** forms when methylbut-2-ene reacts with KMnO_4 .

State the essential conditions for this reaction.

[1]

40. MAR 24/P22/Q2(d)(i)

Context

2 Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

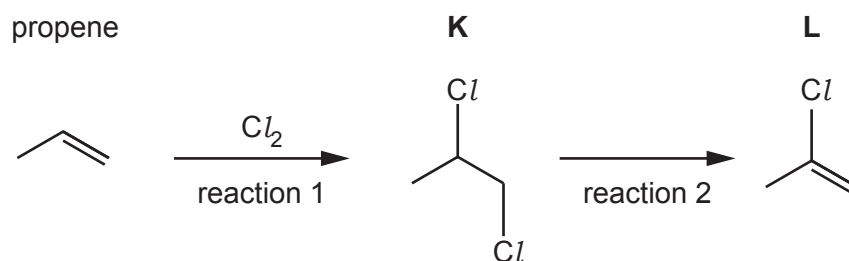
Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

Context: part (d) (introduction)

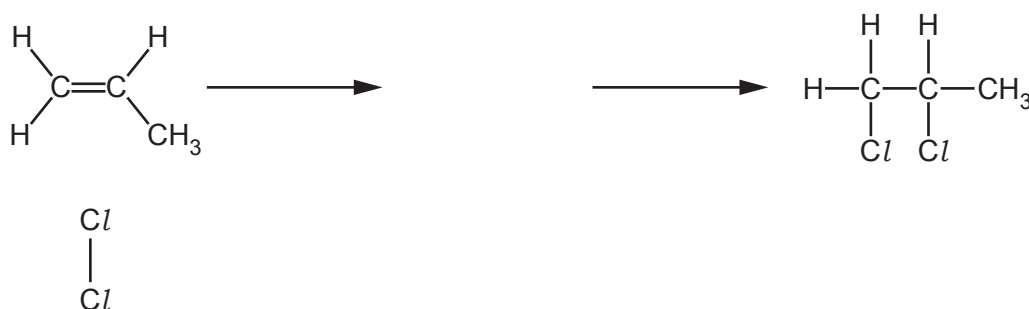
(d) Cl_2 reacts readily with propene to form **K**, 1,2-dichloropropane.

K can be used to form **L**.

**Fig. 2.1**

(i) Complete Fig. 2.2 to show the mechanism for the reaction of Cl_2 with propene in reaction 1.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

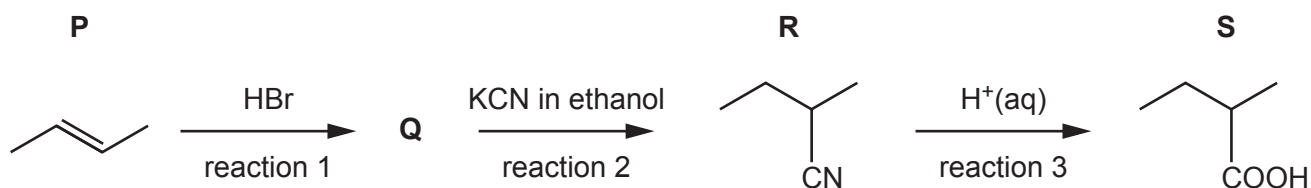
**Fig. 2.2**

[4]

41. MAR 24/P22/Q4(b)(i)

Context

4 Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

**Fig. 4.1**

(b) (i) Give the structural formula of **Q**.

[1]

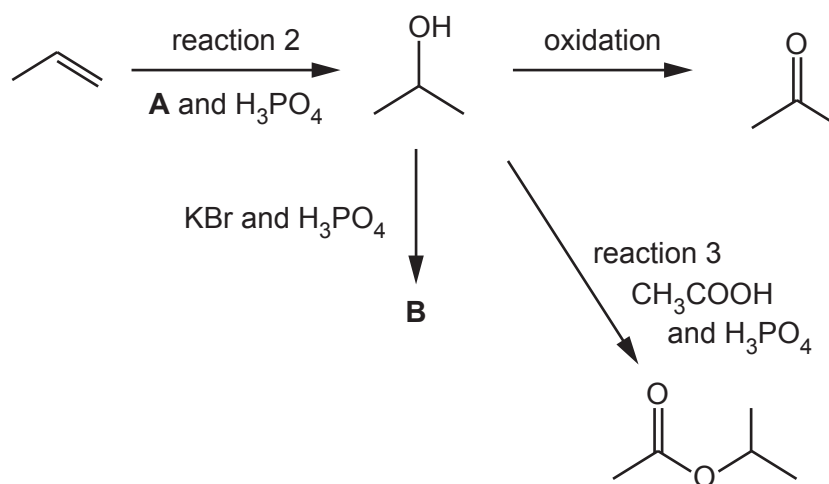
42. O/N 23/P22/Q3(c)(i)

Context

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

Context: part (c) (introduction)

(c) Fig. 3.1 shows a reaction scheme that involves H_3PO_4 in several reactions.

**Fig. 3.1**

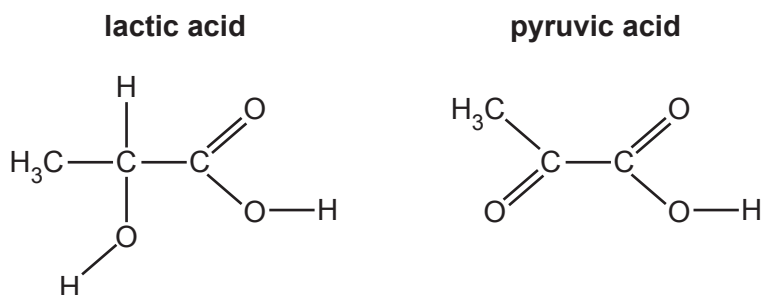
(i) Identify **A**, which reacts with propene in the presence of H_3PO_4 in reaction 2.

[1]

43. O/N 23/P22/Q4(b)(i)

Context

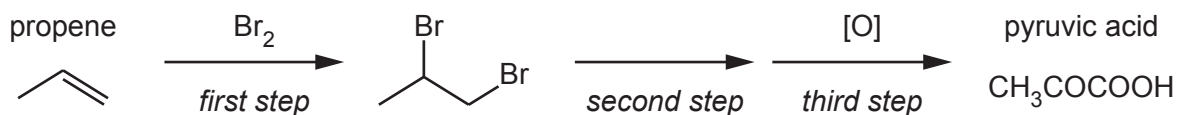
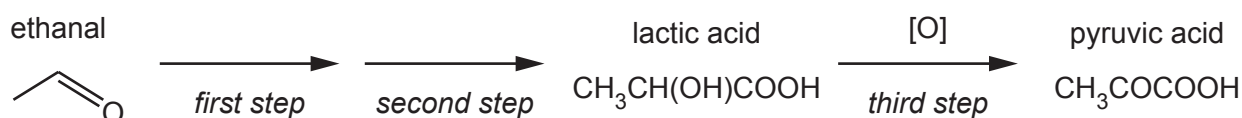
- 4 Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, and pyruvic acid, CH_3COCOOH , both contain two functional groups.

**Fig. 4.1**

Context: part (b) (introduction)

- (b) Two possible syntheses of pyruvic acid are shown in Fig. 4.3 and Fig. 4.4.

Each synthesis has a total of three steps.

**Fig. 4.3****Fig. 4.4**

- (i) Complete the diagram in Fig. 4.5 to show the mechanism for the reaction of propene with Br_2 .

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Fig. 4.5**

44. M/J 23/P21/Q3(d)

Context

3 Chlorine is a very reactive element.**(d)** Chlorine reacts with methane in a series of reactions to produce chloroalkanes.**(i)** State the conditions required for chlorine to react with methane.

[1]

(ii) One of the products of the reaction is CH_2Cl_2 which reacts further to produce CHCl_3 .Complete Table 3.2 to show details of the mechanism that forms CHCl_3 from CH_2Cl_2 .**Table 3.2**

name of step	equation
initiation	
propagation	$\text{CH}_2\text{Cl}_2 + \text{Cl}\cdot \rightarrow$
termination	 $\rightarrow \text{CHCl}_3$

[3]

45. M/J 23/P21/Q5(a)**5 (a)** Molecule **M** is present in petrol, a fuel used in cars. **M** is a saturated, non-cyclic hydrocarbon. **M** contains eight carbon atoms.**(i)** Construct an equation for the complete combustion of **M**.

[2]

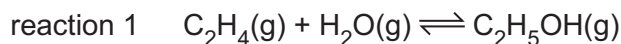
(ii) Describe how the composition of products differs when incomplete combustion of **M** occurs.

[2]

46. M/J 23/P22/Q4(c)(ii)-(iii)

Context

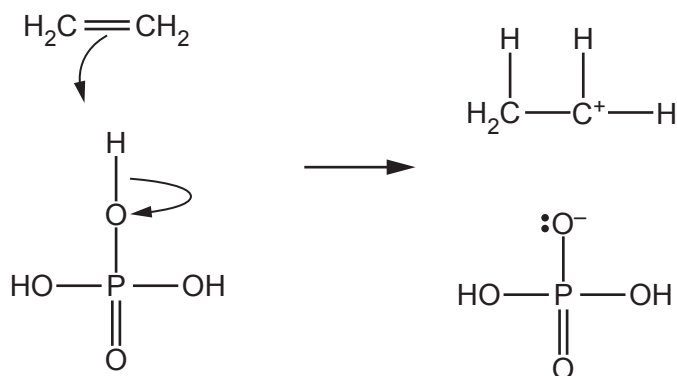
4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



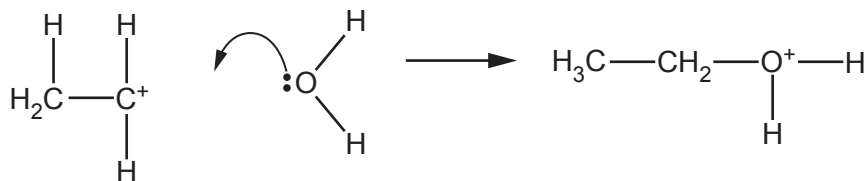
Context: part (c) (introduction)

(c) The mechanism for reaction 1 can be described in three steps. Steps 1 and 2 for reaction 1 are shown in Fig. 4.1.

step 1



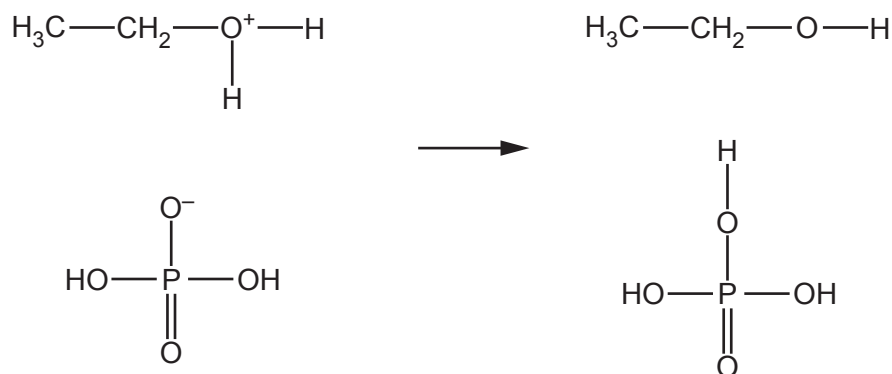
step 2

**Fig. 4.1**

(ii) Identify the species that behaves as an electrophile in step 2 in Fig. 4.1. Explain your answer.

[1]

(iii) Complete Fig. 4.2 to show the mechanism for step 3 of reaction 1. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

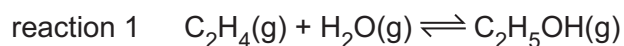
**Fig. 4.2**

[2]

47. M/J 23/P22/Q4(c)(vi)

Context

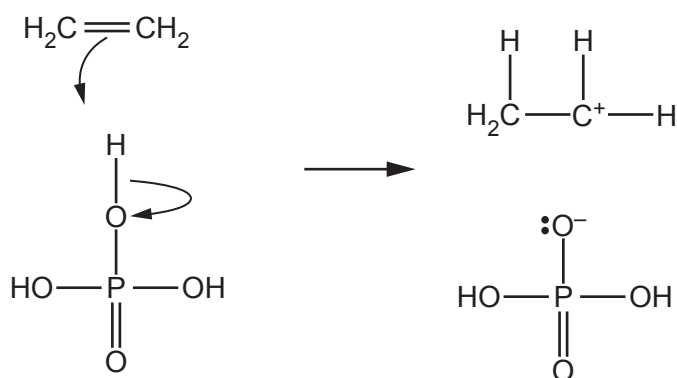
4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



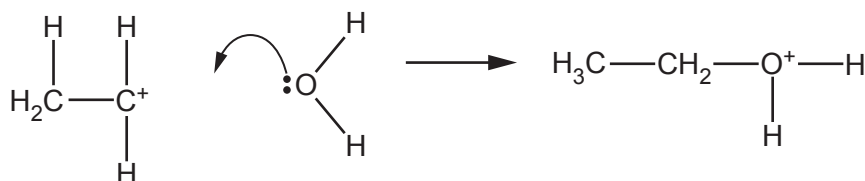
Context: part (c) (introduction)

(c) The mechanism for reaction 1 can be described in three steps. Steps 1 and 2 for reaction 1 are shown in Fig. 4.1.

step 1



step 2

**Fig. 4.1**

(vi) Propene also reacts with steam. A mixture of organic products is produced.

Explain why propan-2-ol is produced in the higher yield.

[2]

48. O/N 22/P21/Q5(a)(i)

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

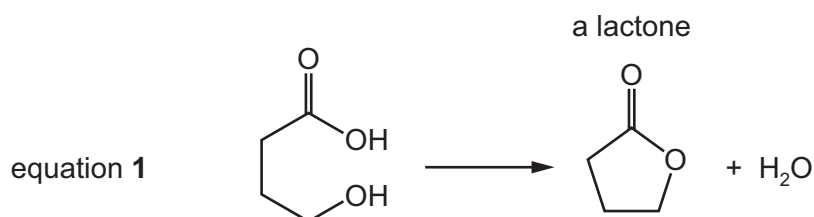


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

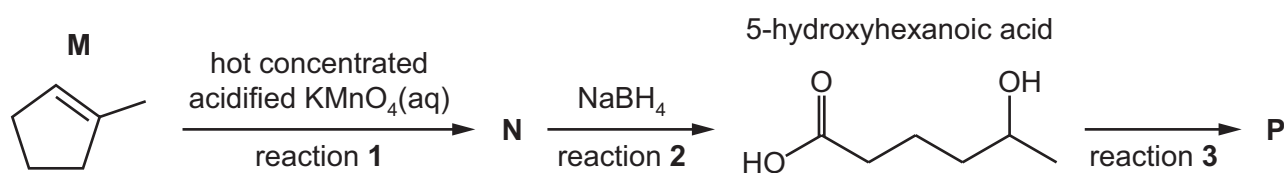


Fig. 5.1

- (a) (i)** **M** reacts with hot concentrated acidified KMnO₄(aq) to form **N**, C₆H₁₀O₃, in reaction 1.

Draw the structure of **N**.

[1]

49. O/N 22/P23/Q5(a)(i)

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

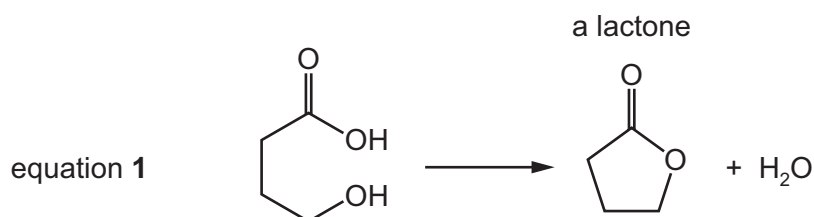
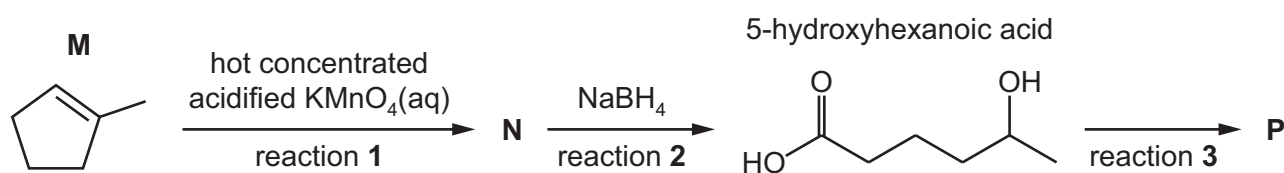


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

**Fig. 5.1**

- (a) (i)** **M** reacts with hot concentrated acidified $\text{KMnO}_4(\text{aq})$ to form **N**, $\text{C}_6\text{H}_{10}\text{O}_3$, in reaction 1.

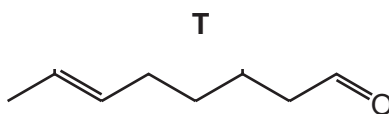
Draw the structure of **N**.

[1]

50. M/J 22/P21/Q3(b)

Context

- 3** Liquids that contain molecules of **T** smell like lemons.

**Fig. 3.1**

- (b)** Two organic products are produced when a sample of **T** is heated under reflux with excess acidified concentrated KMnO_4 .

Draw the structure of the two organic products, from this reaction, in the boxes.

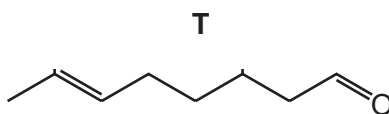


[2]

51. M/J 22/P21/Q3(c)(ii)-(iii)

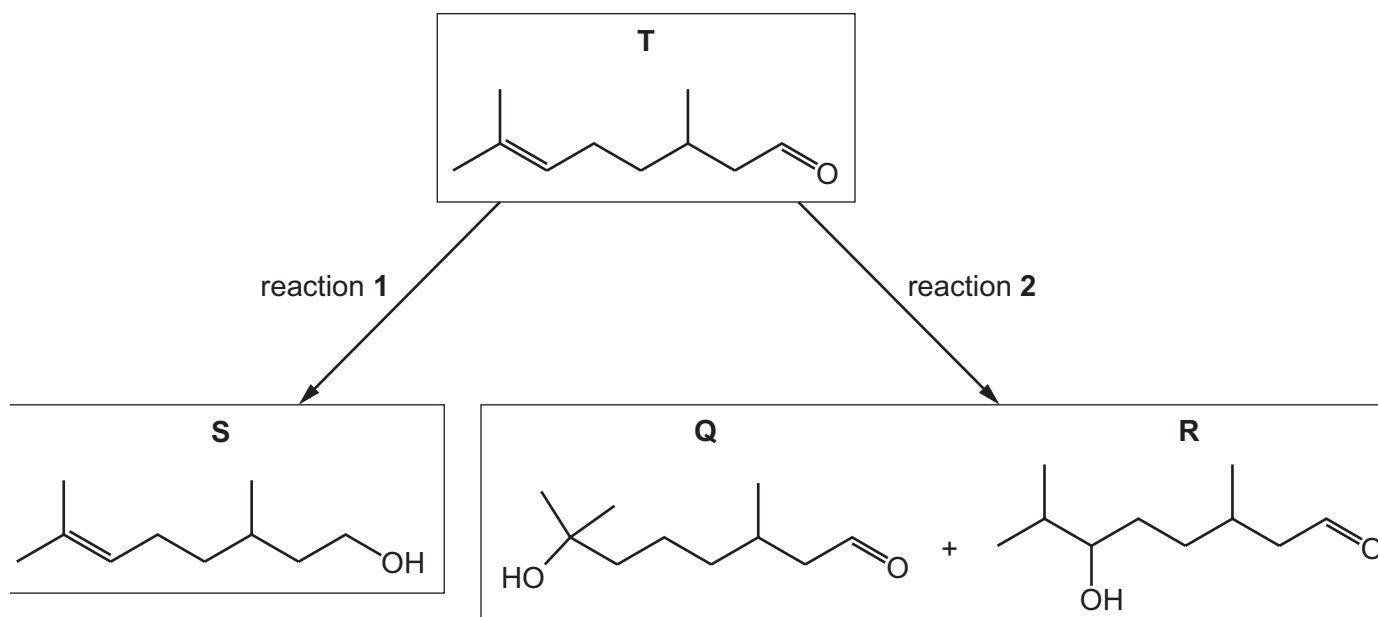
Context

- 3** Liquids that contain molecules of **T** smell like lemons.

**Fig. 3.1**

Context: part (c) (introduction)

- (c)** Fig. 3.2 shows two reactions of **T**.

**Fig. 3.2**

- (ii)** Identify the reagent and conditions needed for reaction 2.

[2]

- (iii)** Suggest which product formed in reaction 2 has a higher yield. Explain your answer.

[3]

52. M/J 22/P23/Q4(a)

- 4 (a) 2-methylpropene reacts with HCl(g) at room temperature. The major organic product is 2-chloro-2-methylpropane.

(i) Complete Fig. 4.1 to show the structure of the intermediate and mechanism for this reaction. Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



Fig. 4.1

[3]

(ii) Explain why, in this reaction, 2-chloro-2-methylpropane is produced at a higher yield than 1-chloro-2-methylpropane.

[2]

53. M/J 22/P23/Q5(a)(i)

Context: part (a) (introduction)

- 5 (a) But-2-ene reacts with KMnO_4 to form organic product, **Y**.
Y does not react with Na_2CO_3 .
A gas is produced when an excess of Na is added to **Y**.

(i) Describe the conditions for the KMnO_4 used in the reaction to form **Y** from but-2-ene.

[1]

54. MAR 22/P22/Q3(e)(i)-(ii)

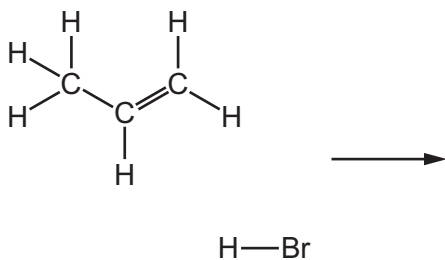
Context

3 The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.

Context: part (e) (introduction)

(e) HBr reacts with propene to form two bromoalkanes, $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ and $(\text{CH}_3)_2\text{CHBr}$.

- (i)** Complete the diagram to show the mechanism of the reaction of HBr and propene to form the major organic product.
Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.
Draw the structures of the intermediate and the major organic product.



[4]

- (ii)** Explain why the two bromoalkanes are **not** produced in equal amounts by this reaction.

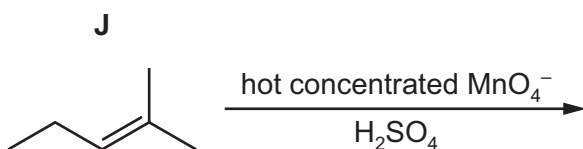
[2]

55. O/N 21/P21/Q3(d)

Context

3 Phosphorus is a reactive Period 3 element.

- (d)** Identify the organic products formed when compound **J**, shown below, is heated with hot concentrated acidified manganate(VII) ions.

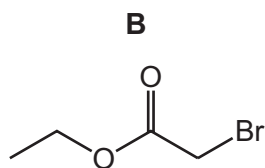


[2]

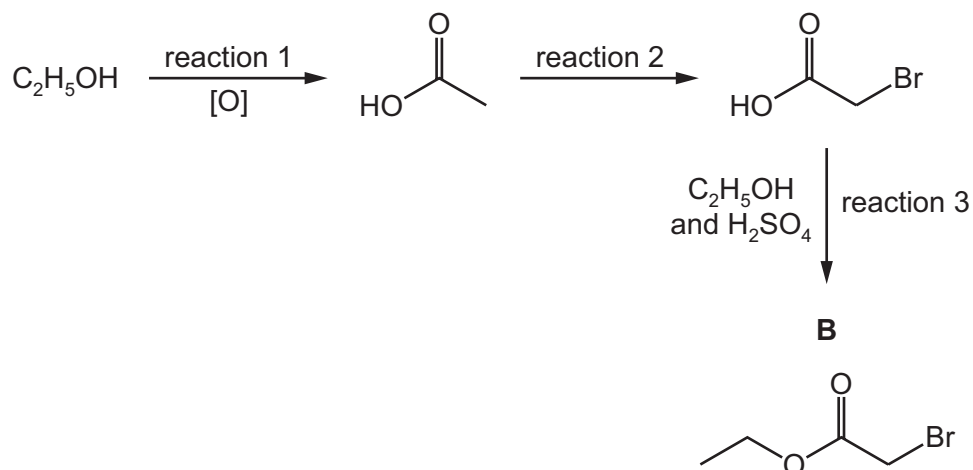
56. O/N 21/P21/Q4(a)(iii)

Context

- 4** Compound **B** is a liquid with a fruity smell.



The reaction scheme shows how **B** can be made from ethanol, $\text{C}_2\text{H}_5\text{OH}$.



- (iii)** Suggest the type of reaction that occurs in reaction 2.

[1]

57. O/N 21/P22/Q1(g)

Context

1 Hydrogen iodide, HI, is a colourless gas at room temperature.

(g) HI(g) reacts with propene, $\text{CH}_3\text{CH}=\text{CH}_2(\text{g})$ to form a mixture of 1-iodopropane and 2-iodopropane.

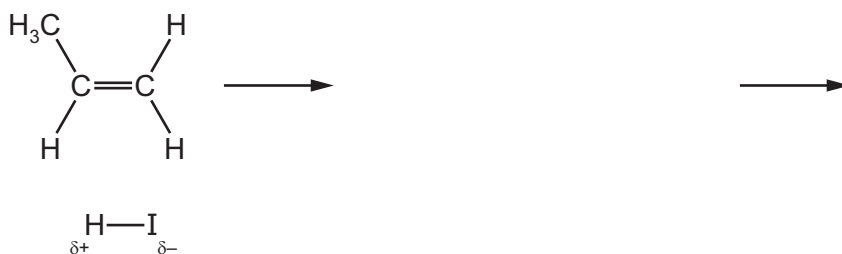
(i) Identify which of 1-iodopropane and 2-iodopropane is the major product of this reaction.

Explain your answer.

[2]

(ii) Complete the diagram to show the mechanism of the reaction between HI and $\text{CH}_3\text{CH}=\text{CH}_2$ that forms the major product identified in **(g)(i)**.

Include curly arrows, lone pairs of electrons and charges as necessary.

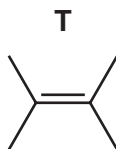


[3]

58. O/N 21/P22/Q3(d)(i)

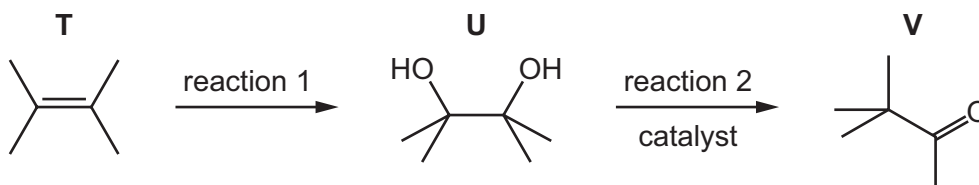
Context

3 Compound **T** is an isomer of C_6H_{12} .



Context: part (d) (introduction)

(d) A reaction scheme starting with **T** is shown. Reaction 2 occurs in the presence of a catalyst; knowledge of the mechanism for this reaction is not required.



(i) Give the reagent(s) and conditions for reaction 1.

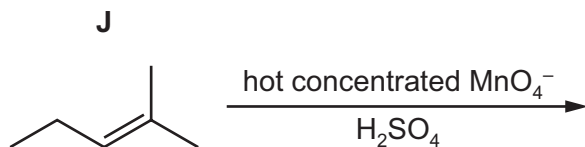
[1]

59. O/N 21/P23/Q3(d)

Context

3 Phosphorus is a reactive Period 3 element.

(d) Identify the organic products formed when compound **J**, shown below, is heated with hot concentrated acidified manganate(VII) ions.

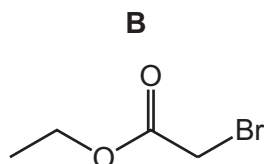


[2]

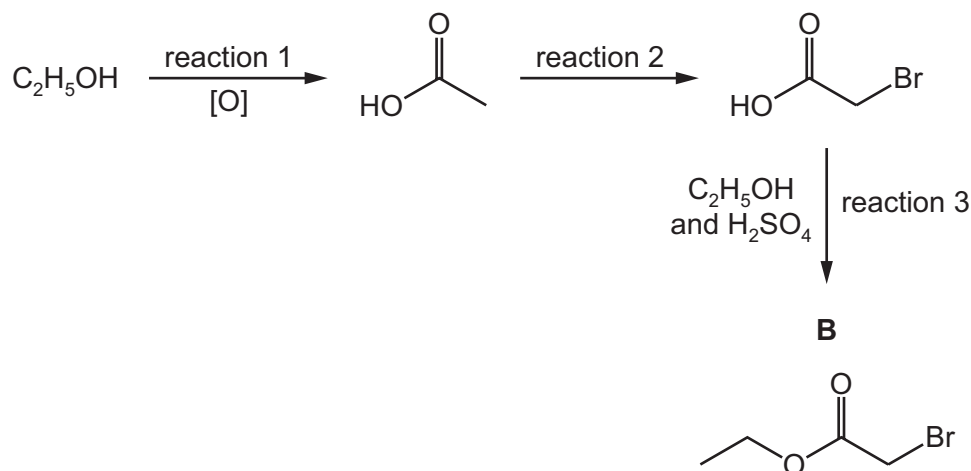
60. O/N 21/P23/Q4(a)(iii)

Context

4 Compound **B** is a liquid with a fruity smell.



The reaction scheme shows how **B** can be made from ethanol, $\text{C}_2\text{H}_5\text{OH}$.



(iii) Suggest the type of reaction that occurs in reaction 2.

[1]

61. M/J 21/P21/Q5(b)

(b) Compound **V** is found in naphtha. It has a molecular formula $C_{10}H_{22}$.

When **V** is heated at high pressure in the absence of air, an equal number of moles of ethene, propene and **W** are made. **W** is a compound made of straight chain, saturated molecules.

(i) Name the process that describes this reaction.

[1]

(ii) Deduce the structure of **W**. Draw its structure below.

[1]

62. M/J 21/P21/Q5(c)(ii)

Context: part (c) (introduction)

(c) Propene is separated from the mixture and heated in air in the presence of a catalyst. Propene is oxidised to **X**, which contains two functional groups.

(ii) Identify a reagent which could be used to show that **X** contains a $C=C$. Include relevant observations.

[2]

63. M/J 21/P21/Q6(a)

Context

6 Propene, C_3H_6 , reacts with H_2O in the presence of an acid catalyst to form an alcohol with molecular formula C_3H_8O .

(a) Name this type of reaction.

[1]

64. M/J 21/P21/Q6(b)

Context

6 Propene, C_3H_6 , reacts with H_2O in the presence of an acid catalyst to form an alcohol with molecular formula C_3H_8O .

(b) Name the catalyst used and state the conditions needed for this reaction to occur.

catalyst

conditions

[2]

65. M/J 21/P21/Q6(d)

Context

- 6** Propene, C_3H_6 , reacts with H_2O in the presence of an acid catalyst to form an alcohol with molecular formula $\text{C}_3\text{H}_8\text{O}$.
- (d)** The reaction of propene, C_3H_6 , with H_2O occurs in a two-step mechanism. In step 1 C_3H_6 reacts with the catalyst, H^+ , to form a carbocation.
- (i)** Draw structures to identify the more stable and less stable carbocations which can form in step 1. Explain your answer.

more stable carbocation	less stable carbocation

explanation

[3]

- (ii) Name the major organic product formed from the reaction of propene, C_3H_6 , with H_2O .

[1]

66. M/J 21/P22/Q4(b)

Context: part (a) (introduction)

- 4 (a) The table shows the structural formulae of four compounds, **A**, **B**, **C** and **D**, with molecular formula C_4H_8 .
- (b) **W** is an alkene with formula C_4H_8 . It reacts with HBr to form two possible carbocations, $CH_3C^+(H)(CH_2CH_3)$ and $H_2C^+CH_2CH_2CH_3$.
- (i) Identify **W** as compound **A**, **B**, **C** or **D**.

[1]

- (ii) Draw the skeletal formula of the major organic product formed when HBr reacts with **W**. Explain why this is the major organic product.

[3]

67. MAR 21/P22/Q2(g)(i)-(ii)

Context

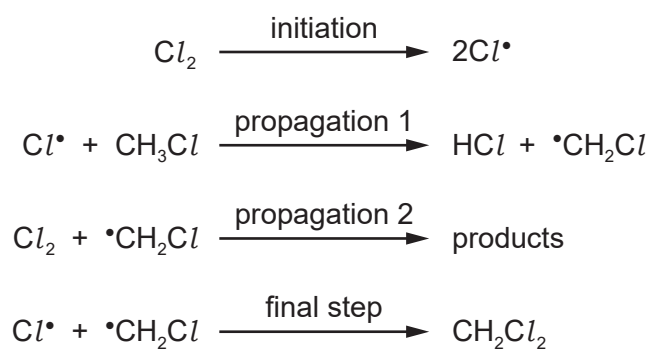
2 Chlorine, Cl_2 , is a reactive yellow-green gas. It is a strong oxidising agent.

Context: part (g) (introduction)

(g) Dichloromethane, CH_2Cl_2 , is widely used as an organic solvent.

CH_2Cl_2 can be prepared by reacting CH_3Cl and Cl_2 at room temperature.

The reaction proceeds via several steps, as shown.



(i) Give the name of the mechanism of this reaction.

[1]

(ii) State the essential condition required for the initiation step to take place.

[1]

68. MAR 21/P22/Q2(g)(iv)-(vi)

Context

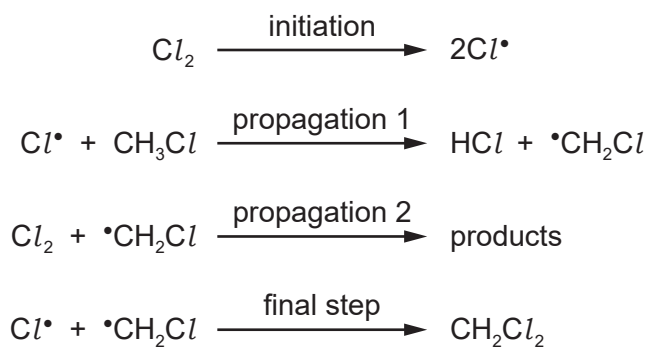
2 Chlorine, Cl_2 , is a reactive yellow-green gas. It is a strong oxidising agent.

Context: part (g) (introduction)

(g) Dichloromethane, CH_2Cl_2 , is widely used as an organic solvent.

CH_2Cl_2 can be prepared by reacting CH_3Cl and Cl_2 at room temperature.

The reaction proceeds via several steps, as shown.



(iv) Identify the products of the step labelled propagation 2.

[1]

(v) Name the type of reaction shown in the final step.

[1]

(vi) Suggest the identity of another organic molecule that is a product of the reaction of CH_3Cl and Cl_2 under the same conditions.

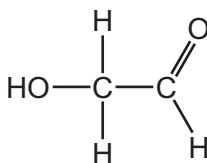
[1]

69. MAR 21/P22/Q4(d)(iii)

Context

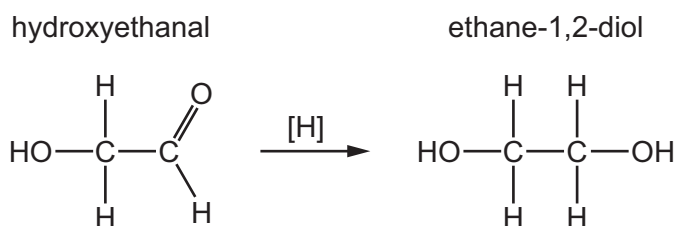
- 4** Hydroxyethanal, HOCH_2CHO , has been observed in dust clouds near the centre of our galaxy.

hydroxyethanal



Context: part (d) (introduction)

- (d)** Hydroxyethanal can be reduced to ethane-1,2-diol, $(\text{CH}_2\text{OH})_2$, as shown.



- (iii)** $(\text{CH}_2\text{OH})_2$ also forms when an alkene **A** reacts with cold, dilute, acidified manganate(VII) ions.

Name **A**.

[1]

70. O/N 20/P21/Q4(b)(ii)

Context

- 4** Iodine is used in many inorganic and organic reactions.

Context: part (b) (introduction)

- (b)** Iodoalkanes contain carbon-iodine bonds.

The simplest iodoalkane is CH_3I .

- (ii)** 1,2-diiodoethane, $\text{CH}_2\text{ICH}_2\text{I}$, can be made by bubbling ethene into liquid iodine.

Fully name the type of mechanism shown in this reaction.

[1]

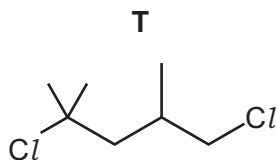
71. O/N 20/P22/Q3(c)(iii)

Context

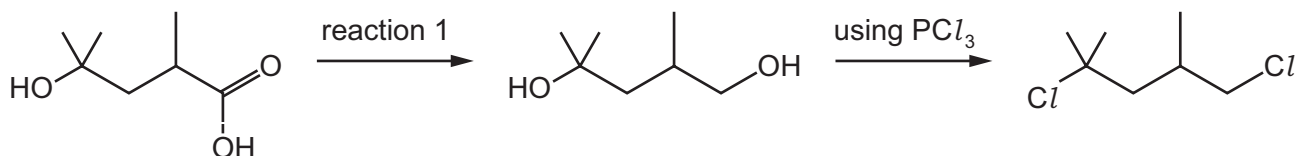
3 PCl_5 , PCl_3 and NCl_3 are halides of Group 15 elements.

Context: part (c) (introduction)

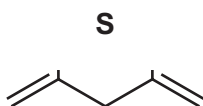
(c) PCl_3 is used to convert alcohols to chloroalkanes, such as compound **T**.



A possible synthesis of **T** is shown.



(iii) **T** is a **minor** product in the reaction of compound **S** with excess HCl .



Draw the structure of the **major** product of the reaction of **S** with excess HCl .

[1]

72. O/N 20/P23/Q4(b)(ii)

Context

4 Iodine is used in many inorganic and organic reactions.

Context: part (b) (introduction)

(b) Iodoalkanes contain carbon-iodine bonds.

The simplest iodoalkane is CH_3I .

(ii) 1,2-diiodoethane, $\text{CH}_2\text{ICH}_2\text{I}$, can be made by bubbling ethene into liquid iodine.

Fully name the type of mechanism shown in this reaction.

[1]

73. M/J 20/P21/Q5(a)(i)-(ii)

Context: part (a) (introduction)

5 (a) Below is a list of species which can react with organic compounds.



(i) From the list, identify a species which can react with ethane.

[1]

(ii) From the list, identify **two** species which can attack the π bond in ethene.

[1]

74. M/J 20/P21/Q5(b)(i)

Context: part (b) (introduction)

(b) Cl(g) can be made from $\text{Cl}_2(\text{g})$.

(i) Describe the conditions required for this process.

[1]

75. M/J 20/P21/Q5(d)

(d) But-1-ene reacts with steam in the presence of concentrated phosphoric acid to form two isomers of molecular formula $\text{C}_4\text{H}_{10}\text{O}$.

Each reaction occurs via a different intermediate ion.

(i) Draw the structure of both intermediate ions.

[2]

(ii) Circle the more stable intermediate ion drawn in **(d)(i)**. Explain your answer.

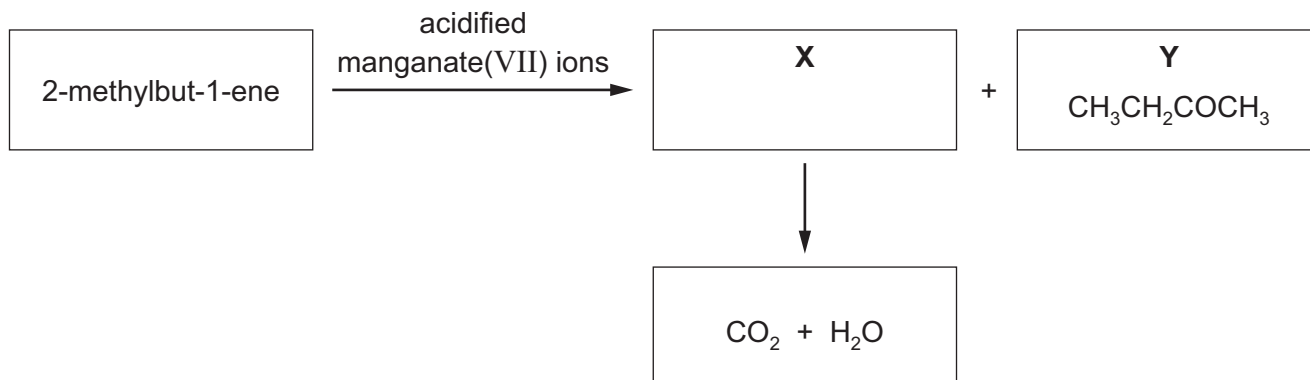
[2]

76. M/J 20/P21/Q6(b)

Context

- 6** 2-methylbut-1-ene reacts with acidified manganate(VII) ions, under specific conditions, to produce two organic compounds **X** and **Y**.

X immediately reacts with the acidified manganate(VII) ions to form carbon dioxide and water. **Y** has the structural formula $\text{CH}_3\text{CH}_2\text{COCH}_3$.



- (b) (i)** State the specific conditions required for the acidified manganate(VII) ions to react with 2-methylbut-1-ene in this way.

[1]

- (ii)** Name the type of reaction occurring to the functional group in 2-methylbut-1-ene in the reaction in **(b)(i)**.

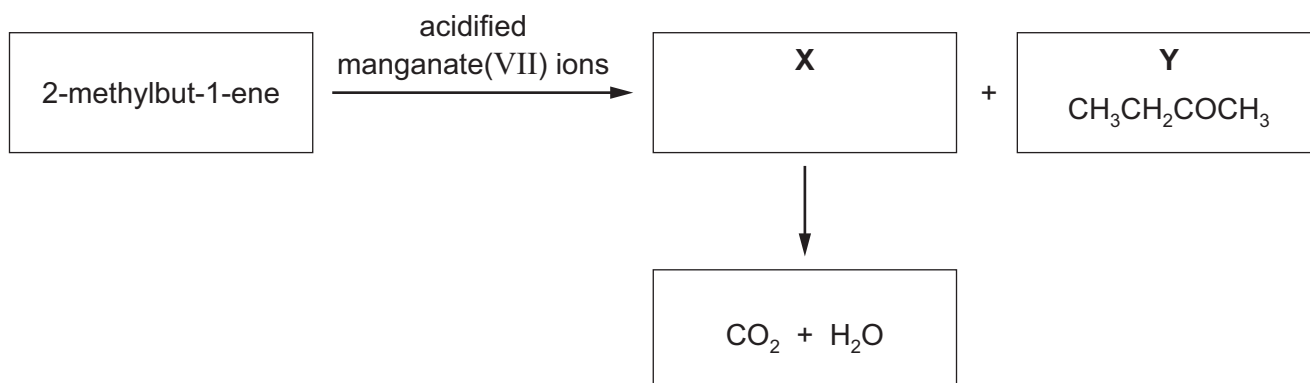
[1]

77. M/J 20/P21/Q6(c)

Context

- 6** 2-methylbut-1-ene reacts with acidified manganate(VII) ions, under specific conditions, to produce two organic compounds **X** and **Y**.

X immediately reacts with the acidified manganate(VII) ions to form carbon dioxide and water. **Y** has the structural formula $\text{CH}_3\text{CH}_2\text{COCH}_3$.



- (c)** Draw the structural formula of **X**.

[1]

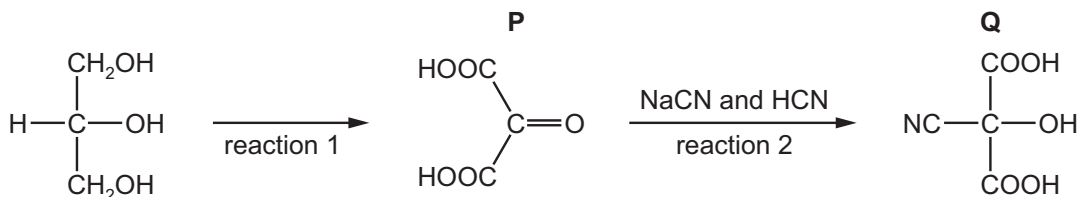
78. MAR 20/P22/Q3(c)(iii)

Context

3 Glycerol, $\text{CH}_2(\text{OH})\text{CH}(\text{OH})\text{CH}_2\text{OH}$, is widely used in the food industry and in pharmaceuticals.

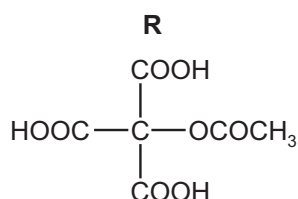
Context: part (a) (introduction)

(a) A series of reactions starting from glycerol is shown.



Context: part (a)(v)

(v) When **Q** is heated with excess aqueous ethanoic acid in the presence of a catalytic amount of sulfuric acid, two reactions take place to form compound **R**.



Identify the two types of reaction that occur.

1

2

[2]

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9701/22/F/M/20

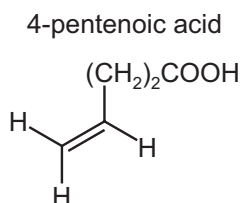
[Turn over]



Context: part (c) (introduction)

(c) Fats are compounds made from glycerol and unsaturated carboxylic acids.

4-pentenoic acid is an example of an unsaturated carboxylic acid.



(iii) Unsaturated acids are often brominated before being added to soft drinks.

Complete the mechanism for the addition of Br_2 to 4-pentenoic acid.

- Include the structures of the intermediate and the product of the reaction.
- Include all charges, partial charges, lone pairs and curly arrows.

In the mechanism, R has been used to represent $(\text{CH}_2)_2\text{COOH}$.





[4]

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[Turn over]

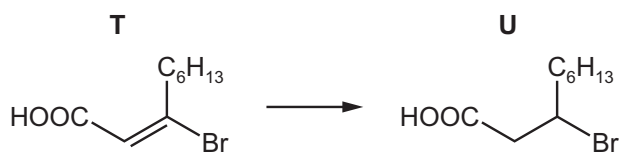
**79. MAR 20/P22/Q3(d)(ii)**

Context

3 Glycerol, $\text{CH}_2(\text{OH})\text{CH}(\text{OH})\text{CH}_2\text{OH}$, is widely used in the food industry and in pharmaceuticals.

Context: part (d) (introduction)

(d) A reaction of another unsaturated carboxylic acid, **T**, is shown.



(ii) Identify the reagent used to convert **T** to **U**.

[1]

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9701/22/F/M/20

**80. O/N 19/P21/Q1(e)**

(e) Under certain conditions, chlorine undergoes a free-radical substitution reaction with ethane.

(i) State the conditions required to initiate this reaction.

[1]

(ii) Write the overall equation for this free-radical substitution reaction.

[1]

81. O/N 19/P21/Q3(a)(i)

Context

3 Crude oil is a natural source of hydrocarbons that are used as fuels.

Context: part (a) (introduction)

(a) Hydrocarbons with low relative molecular mass, M_r , are used as fuels in industry, in the home and for transport.

There is a high demand for the hydrocarbons with low M_r .

(i) Name the process by which long-chain hydrocarbons are broken down into shorter-chain hydrocarbons.

[1]

82. O/N 19/P21/Q3(a)(iii)

Context

- 3** Crude oil is a natural source of hydrocarbons that are used as fuels.

Context: part (a) (introduction)

- (a)** Hydrocarbons with low relative molecular mass, M_r , are used as fuels in industry, in the home and for transport.

There is a high demand for the hydrocarbons with low M_r .

- (iii)** Incomplete combustion of hydrocarbons can release carbon monoxide, CO, into the atmosphere.

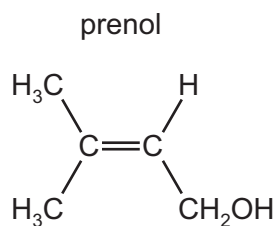
Write an equation for the formation of CO from the incomplete combustion of butene, C_4H_8 .

[1]

83. O/N 19/P22/Q4(c)(i)

Context

- 4** Prenol is a naturally occurring organic molecule found in many fruits. It contains both an alkene and an alcohol functional group.



- (c) (i)** Prenol contains an alkene functional group.

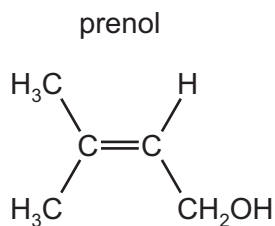
Describe a chemical test to confirm the presence of an alkene functional group. Give the result of the test.

[1]

84. O/N 19/P22/Q4(e)(iv)

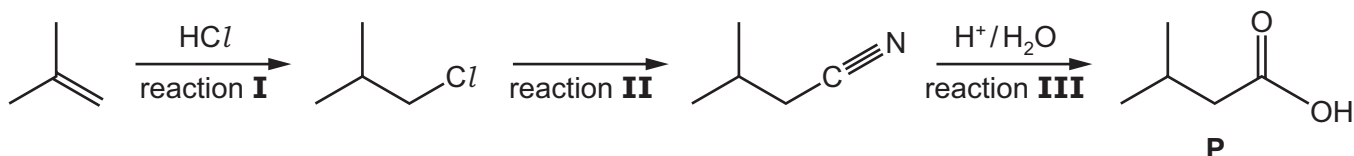
Context

- 4** Prenol is a naturally occurring organic molecule found in many fruits. It contains both an alkene and an alcohol functional group.



Context: part (e) (introduction)

- (e)** **P** can be produced as shown.



- (iv)** The yield of reaction **I** is very low.

Explain why.

[2]

85. O/N 19/P23/Q1(e)

- (e)** Under certain conditions, chlorine undergoes a free-radical substitution reaction with ethane.

- (i)** State the conditions required to initiate this reaction.

[1]

- (ii)** Write the overall equation for this free-radical substitution reaction.

[1]

86. O/N 19/P23/Q3(a)(i)

Context

- 3** Crude oil is a natural source of hydrocarbons that are used as fuels.

Context: part (a) (introduction)

- (a)** Hydrocarbons with low relative molecular mass, M_r , are used as fuels in industry, in the home and for transport.

There is a high demand for the hydrocarbons with low M_r .

- (i)** Name the process by which long-chain hydrocarbons are broken down into shorter-chain hydrocarbons.

[1]

87. O/N 19/P23/Q3(a)(iii)

Context

- 3** Crude oil is a natural source of hydrocarbons that are used as fuels.

Context: part (a) (introduction)

- (a)** Hydrocarbons with low relative molecular mass, M_r , are used as fuels in industry, in the home and for transport.

There is a high demand for the hydrocarbons with low M_r .

- (iii)** Incomplete combustion of hydrocarbons can release carbon monoxide, CO, into the atmosphere.

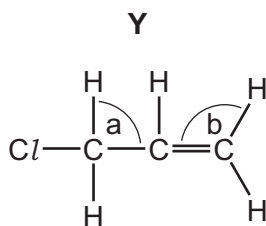
Write an equation for the formation of CO from the incomplete combustion of butene, C_4H_8 .

[1]

88. M/J 19/P21/Q4(c)(ii)

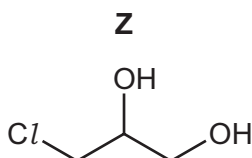
Context

- 4** The structure of compound **Y** is shown.



Context: part (c) (introduction)

- (c)** When **Y** reacts with cold, dilute, acidified manganate(VII) ions, compound **Z** is produced.



- (ii)** Name the type of reaction occurring when **Y** is converted into **Z**.

[1]

89. M/J 19/P22/Q4(b)

Context

- 4** There are many different types of aliphatic and aromatic hydrocarbons.

- (b)** When alkanes are heated to high temperatures, in the absence of air, the molecules can break into smaller molecules.

- (i)** Identify the type of reaction occurring.

[1]

- (ii)** Write an equation which describes the reaction occurring when heptane, C_7H_{16} , is heated in the absence of air, to form hexane, butane and ethene only.

[1]

90. M/J 19/P22/Q4(d)(iii)-(iv)

Context

4 There are many different types of aliphatic and aromatic hydrocarbons.

Context: part (d) (introduction)

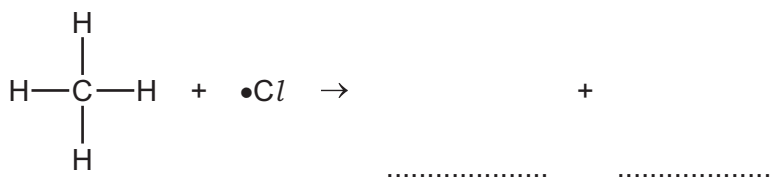
(d) The table compares the reactivity of alkanes and alkenes with chlorine.

	alkanes	alkenes
name of the type of reaction with chlorine	substitution	addition and substitution
name of the type of reacting species	free radical	electrophile and free radical

(iii) Name the stage of the reaction mechanism which occurs when a methane molecule reacts with a chlorine free radical.

[1]

(iv) Complete the equation for the reaction which occurs when a methane molecule reacts with a chlorine free radical.



[1]

91. M/J 19/P23/Q6(a)(i)-(ii)

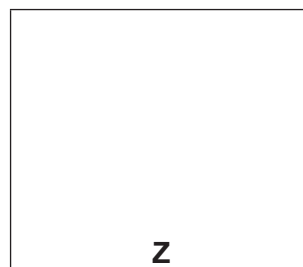
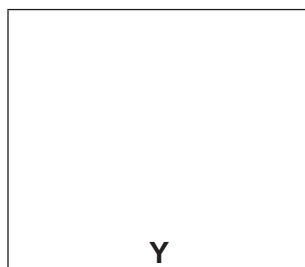
Context: part (a) (introduction)

6 (a) Three alkenes, **X**, **Y** and **Z**, have the same molecular formula.**(i)** Describe what is seen when aqueous bromine is added to **X**.

[1]

X, **Y** and **Z** are reacted separately with hot, concentrated, acidified manganate(VII) ions until no further reaction occurs. The carbon-containing products are shown in the table.

alkene	carbon-containing products
X	$\text{CO}_2 + (\text{CH}_3)_2\text{CO}$
Y	$\text{CO}_2 + \text{CH}_3\text{CH}_2\text{CO}_2\text{H}$
Z	$\text{CH}_3\text{CO}_2\text{H}$

(ii) Draw the structures of **X**, **Y** and **Z**.

[3]

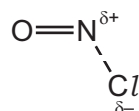
92. MAR 19/P22/Q1(e)

Context

1 Nitrogen, N_2 , is the most abundant gas in the Earth's atmosphere and is very unreactive.

(e) Nitrosyl chloride, $NOCl$, is a reactive gas that is sometimes formed when NO reacts with Cl_2 .

nitrosyl chloride



$NOCl$ is a strong electrophile and readily undergoes an addition reaction with alkenes.

Complete the diagram to show the mechanism of the electrophilic addition reaction of $NOCl$ with ethene.

Include all necessary charges, lone pairs and curly arrows, and the structure of the organic intermediate.



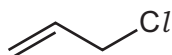
[2]

93. MAR 19/P22/Q4(b)(i)-(iii)

Context

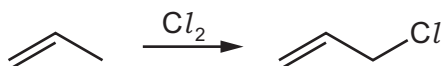
- 4** Allyl chloride is an important chemical used in the manufacture of plastics, pharmaceuticals and pesticides.

allyl chloride



Context: part (b) (introduction)

- (b)** Allyl chloride can be produced by many different methods. The most common method is chlorination of propene which proceeds via a free-radical substitution mechanism.



- (i)** The initiation step in this reaction is the formation of chlorine radicals ($Cl^\bullet$) from Cl_2 molecules.

State the conditions required to initiate this reaction.

[1]

- (ii)** The propenyl radical, $CH_2=CHCH_2^\bullet$, is formed in the first propagation step of the reaction.

Write an equation to show the formation of $CH_2=CHCH_2^\bullet$ in this propagation step.

[1]

- (iii)** Explain why the free-radical substitution reaction gives a low yield of allyl chloride.

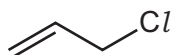
[1]

94. MAR 19/P22/Q4(c)(i)

Context

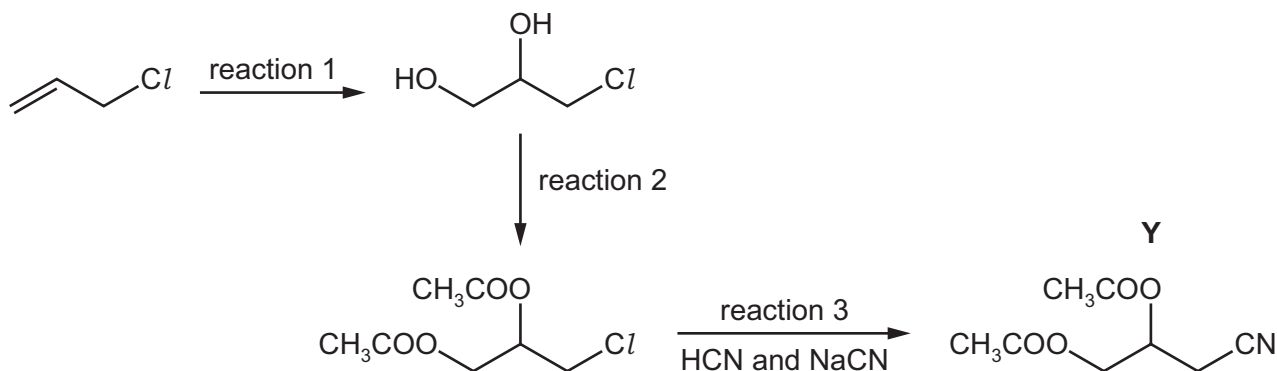
- 4 Allyl chloride is an important chemical used in the manufacture of plastics, pharmaceuticals and pesticides.

allyl chloride



Context: part (c) (introduction)

(c) A series of reactions starting from allyl chloride is shown.



- (i) Suggest a reagent that can be used in reaction 1.

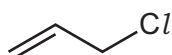
[1]

95. MAR 19/P22/Q4(d)

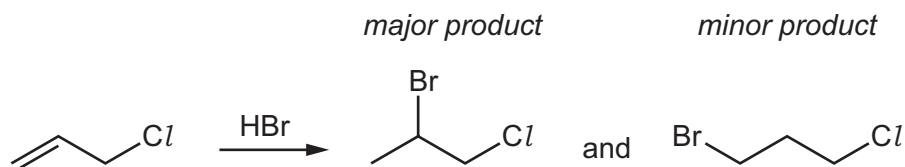
Context

- 4 Allyl chloride is an important chemical used in the manufacture of plastics, pharmaceuticals and pesticides.

allyl chloride



- (d) 2-bromo-1-chloropropane, $\text{CH}_3\text{CHBrCH}_2\text{Cl}$, is the major product of the reaction of allyl chloride with HBr .

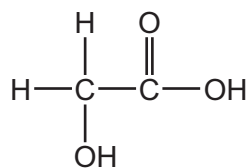


Explain why 2-bromo-1-chloropropane is the major product of this reaction.

[2]

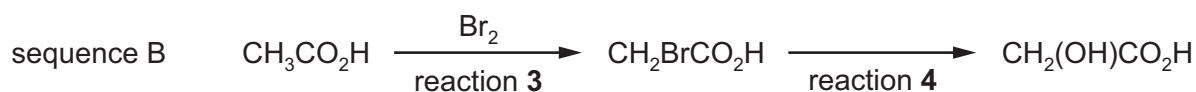
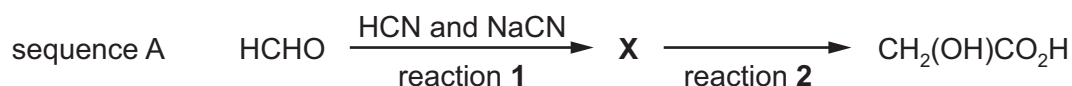
96. O/N 18/P21/Q4(b)(iii)-(iv)

Context

4 The structure of glycolic acid is shown.

glycolic acid

Context: part (b) (introduction)

(b) Two reaction sequences to make glycolic acid are shown.**(iii)** Name the mechanism of reaction 3.

[1]

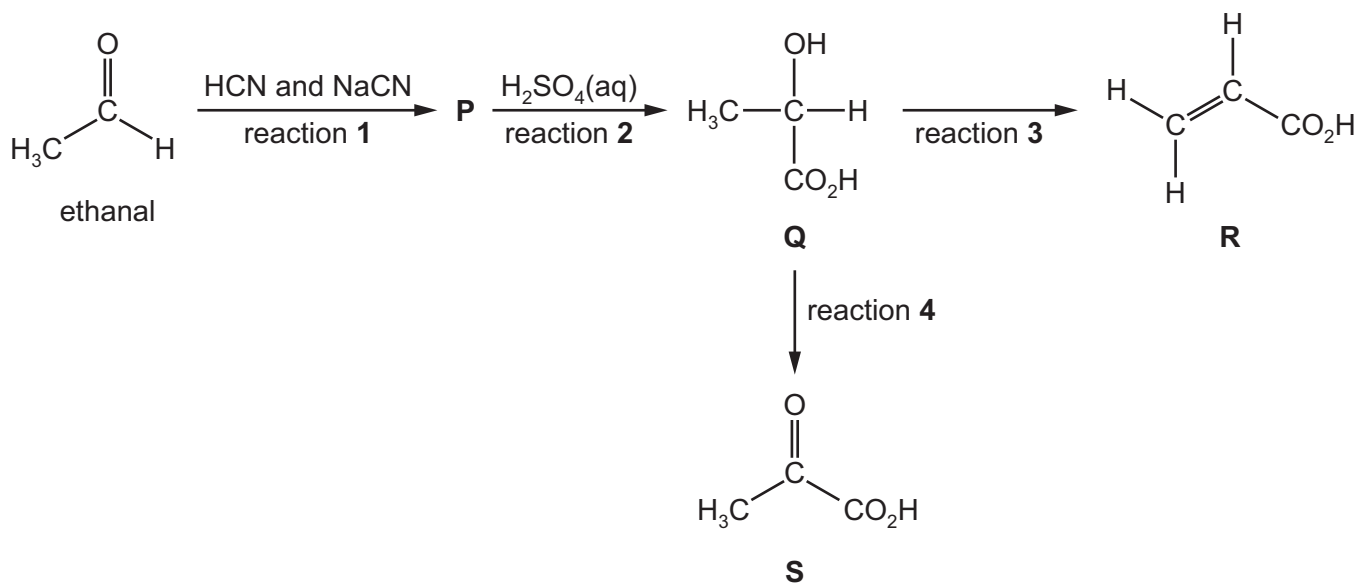
(iv) Suggest the essential condition for reaction 3.

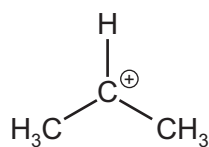
[1]

97. O/N 18/P22/Q4(d)

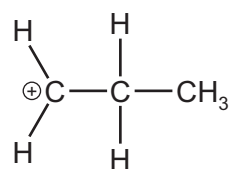
also Unit 15 (1 mark); this unit 5 marks

Context

4 The diagram shows a reaction sequence starting from ethanal.**(d)** Compound **Z**, $\text{H}_2\text{C}=\text{CHCH}_3$, is produced from **R**.**Z** can be used in a two-step process to produce 2-aminopropane.**(i)** In the first step, **Z** reacts with HBr to form two products. The structure of the product depends on which intermediate is formed, intermediate **I** or intermediate **II**.



intermediate I



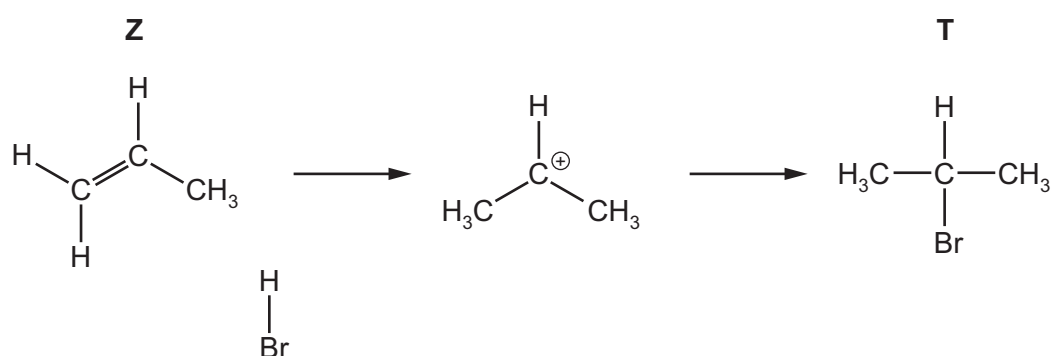
intermediate II

Explain why intermediate I is more likely to form than intermediate II.

[2]

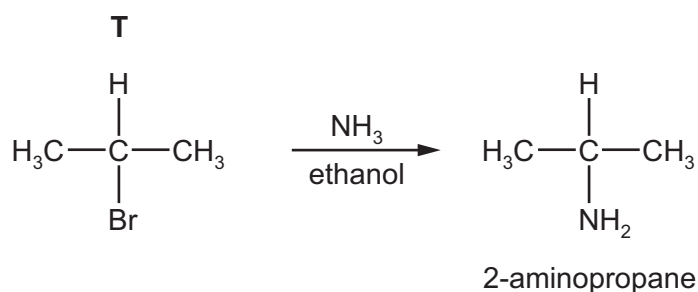
- (ii) When intermediate I forms, the product of the first step is T.

Complete the diagram to show the mechanism for the conversion of Z to T. Include all relevant charges, partial charges, curly arrows and lone pairs.



[3]

- (iii) T can then be converted to 2-aminopropane.



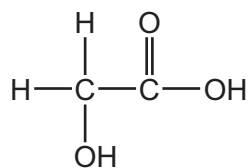
Name the mechanism for this conversion.

[1]

98. O/N 18/P23/Q4(b)(iii)-(iv)

Context

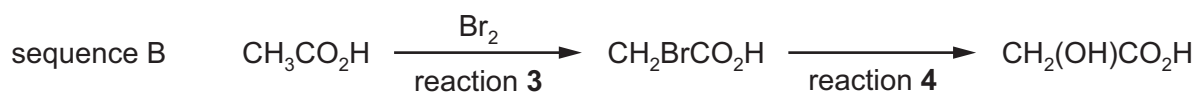
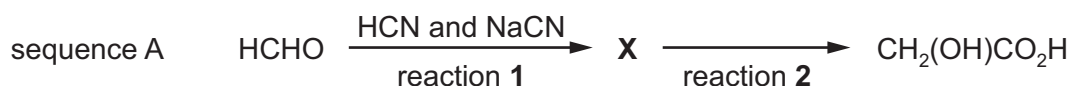
4 The structure of glycolic acid is shown.



glycolic acid

Context: part (b) (introduction)

(b) Two reaction sequences to make glycolic acid are shown.



(iii) Name the mechanism of reaction 3.

[1]

(iv) Suggest the essential condition for reaction 3.

[1]

99. M/J 18/P21/Q2(b)

Context

2 Crude oil is a complex mixture of hydrocarbon molecules.

The hydrocarbon molecules in crude oil are separated by fractional distillation. Fractional distillation is used because the different hydrocarbon molecules in crude oil have different boiling points.

(b) Some of the hydrocarbon molecules obtained from crude oil are processed further by cracking.

Suggest why some hydrocarbon molecules are processed further by cracking.

[1]

100. M/J 18/P21/Q2(c)(i)

Context

- 2** Crude oil is a complex mixture of hydrocarbon molecules.

The hydrocarbon molecules in crude oil are separated by fractional distillation. Fractional distillation is used because the different hydrocarbon molecules in crude oil have different boiling points.

Context: part (c) (introduction)

- (c)** Cracking one mole of dodecane, $C_{12}H_{26}$, produces two moles of ethene and one mole of another hydrocarbon molecule.

- (i)** Write the equation for this cracking reaction.

[1]

The ethene can be used in the production of poly(ethene).

101. M/J 18/P22/Q3(a)

Context

- 3** Most vehicle fuels contain hydrocarbons obtained from crude oil.

- (a) (i)** State the name of the type of reaction that hydrocarbons undergo when being used as fuels.

[1]

- (ii)** Write an equation for the reaction of octane, C_8H_{18} , as a fuel, as in **(a)(i)**.

[2]

102. MAR 18/P22/Q2(b)(ii)

Context

- 2** Carbon and silicon are elements in Group 14.

Context: part (b) (introduction)

- (b)** C_{60} forms hydrocarbons with similar chemical properties to those of alkenes. One such hydrocarbon is $C_{60}H_{18}$.

- (ii)** Describe a test to indicate the presence of double bonds between carbon atoms in $C_{60}H_{18}$. Give the result of the test.

test

result

[2]

103. MAR 18/P22/Q2(c)(i)

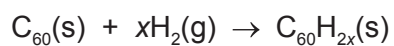
Context

2 Carbon and silicon are elements in Group 14.

Context: part (c) (introduction)

(c) 0.144 g of C_{60} was placed in a 100 cm^3 container of hydrogen gas at 20°C and $1.00 \times 10^5\text{ Pa}$.The container was heated to make the C_{60} and hydrogen gas react.

The reaction occurred as shown in the equation.

After the reaction, the container was allowed to cool to 20°C . The pressure decreased to $2.21 \times 10^4\text{ Pa}$. All of the C_{60} had reacted.**(i)** Name the type of reaction that occurred.

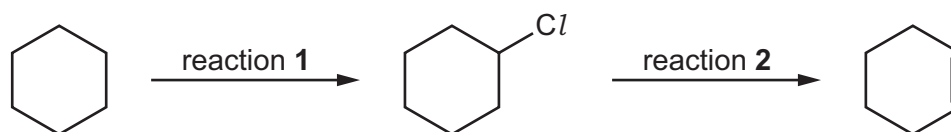
[1]

104. MAR 18/P22/Q4(a)

Context

4 Cyclohexane is a colourless liquid used in industry to produce synthetic fibres.

A reaction scheme involving cyclohexane is shown.

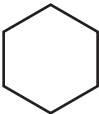
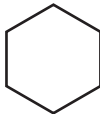
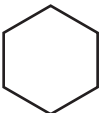
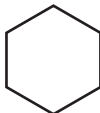
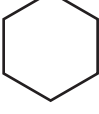


(a) Reaction 1 involves a free radical substitution mechanism.

(i) State the essential condition required for reaction 1 to occur.

[1]

(ii) Complete the table to give details of the mechanism in reaction 1.

name of step	reaction
.....	$\text{Cl}_2 \longrightarrow 2\text{Cl}\cdot$
propagation	 $+ \text{Cl}\cdot \longrightarrow$  $+ \dots\dots\dots$
	 $+ \text{Cl}_2 \longrightarrow$  $+ \text{Cl}\cdot$
termination	 $+ \text{Cl}\cdot \longrightarrow$

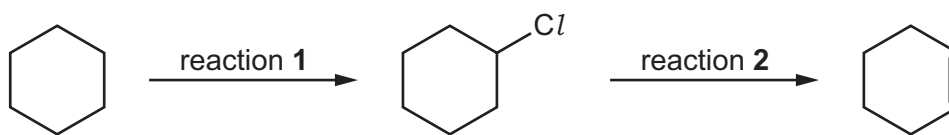
[4]

105. MAR 18/P22/Q4(c)(i)

Context

4 Cyclohexane is a colourless liquid used in industry to produce synthetic fibres.

A reaction scheme involving cyclohexane is shown.



Context: part (c) (introduction)

(c) The product of reaction 2 is cyclohexene.

Cyclohexene can be converted into adipic acid (hexanedioic acid), $\text{HO}_2\text{C}(\text{CH}_2)_4\text{CO}_2\text{H}$.

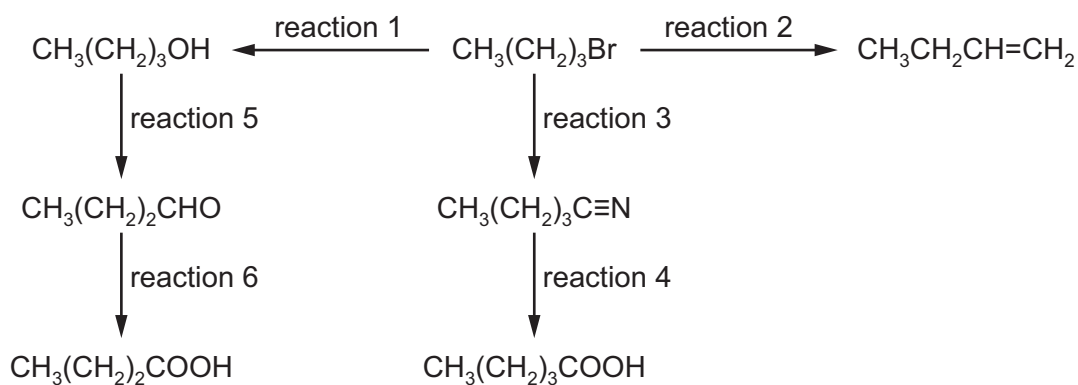
(i) Identify the reagents and conditions for the conversion of cyclohexene into adipic acid.

[2]

106. O/N 17/P21/Q3(d)(iii)

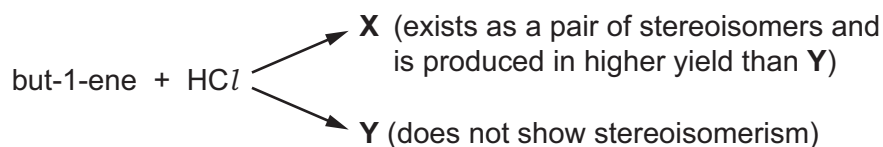
Context

3 Some reactions based on 1-bromobutane, $\text{CH}_3(\text{CH}_2)_3\text{Br}$, are shown.



Context: part (d) (introduction)

(d) The product of reaction 2, but-1-ene, does **not** show stereoisomerism. However, but-1-ene reacts with HCl to form a mixture of structural isomers **X** and **Y**.



(iii) Name **X** and **Y**.

X

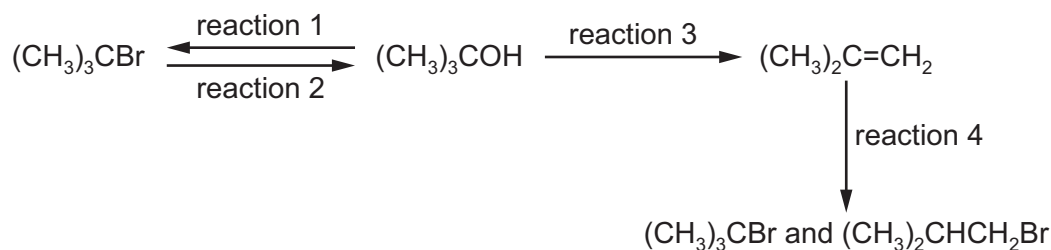
Y

[2]

107. O/N 17/P22/Q4(e)

Context

4 Some reactions are shown, based on methylpropan-2-ol, $(\text{CH}_3)_3\text{COH}$.



(e) Name the two products of reaction 4.

name of $(\text{CH}_3)_3\text{CBr}$

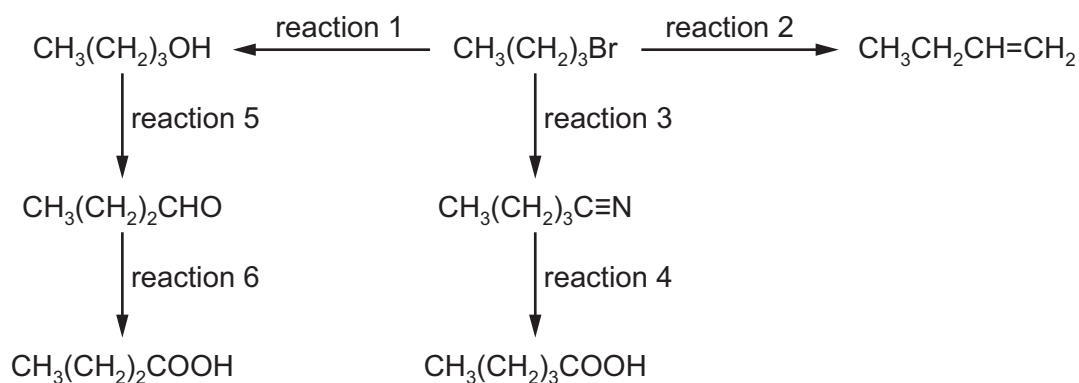
name of $(\text{CH}_3)_2\text{CHCH}_2\text{Br}$

[2]

108. O/N 17/P23/Q3(d)(iii)

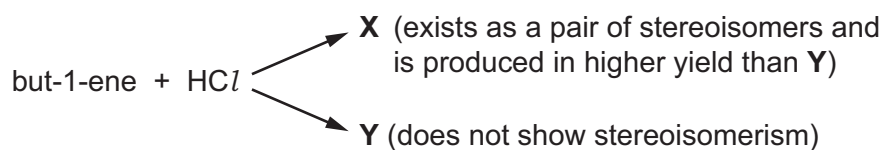
Context

3 Some reactions based on 1-bromobutane, $\text{CH}_3(\text{CH}_2)_3\text{Br}$, are shown.



Context: part (d) (introduction)

(d) The product of reaction 2, but-1-ene, does **not** show stereoisomerism. However, but-1-ene reacts with HCl to form a mixture of structural isomers **X** and **Y**.



(iii) Name **X** and **Y**.

X

Y

[2]

109. M/J 17/P21/Q4(b)

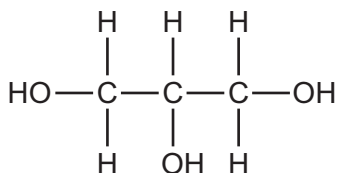
Context

4 **P**, **Q** and **R** all have the molecular formula C_3H_6O . They are all structural isomers of each other.

P and **Q** each contain an oxygen atom bonded directly to a carbon atom that is sp^2 hybridised.
R contains an oxygen atom bonded directly to a carbon atom that is sp^3 hybridised.

(b) **R** contains two different functional groups, one of which is an alkene group.

R reacts with cold, dilute, acidified manganate(VII) ions to form propane-1,2,3-triol.



propane-1,2,3-triol

(i) Give the displayed formula of **R**.

[1]

(ii) State the type of reaction and what you would observe when **R** reacts with bromine water.

[2]

(iii) Draw the structure of the product formed when **R** reacts with bromine water.

[1]

(iv) Identify the gaseous product formed when **R** reacts with hot, concentrated, acidified manganate(VII) ions.

[1]

110. M/J 17/P22/Q4(a)(v)-(vi)

Context: part (a) (introduction)

4 **(a)** The hydrocarbons **A**, C_4H_{10} , and **B**, C_4H_8 , are both unbranched.

A does **not** decolourise bromine.

B decolourises bromine and shows geometrical isomerism.

(v) Draw the mechanism of the reaction of **B** with bromine, Br_2 .
Include all necessary charges, dipoles, lone pairs and curly arrows.

[4]

(vi) Explain the origin of the dipole on Br_2 in this mechanism.

[1]

111. M/J 17/P23/Q4(a)

Context

- 4** **A**, **B** and **C** all have the formula C_4H_8 . They all decolourise bromine and are structural isomers of each other.

(a) State the name of the process by which **A**, **B** and **C** could be obtained from $C_{10}H_{22}$.

[1]

112. M/J 17/P23/Q4(c)(ii)

Context

- 4** **A**, **B** and **C** all have the formula C_4H_8 . They all decolourise bromine and are structural isomers of each other.

Context: part (c) (introduction)

(c) Only **A** shows geometrical isomerism.

- (ii) Draw the displayed formula of **A** and use it to show the mechanism of the reaction of **A** with HBr. Include all necessary charges, dipoles, lone pairs and curly arrows.

[4]

113. M/J 17/P23/Q4(d)(iv)

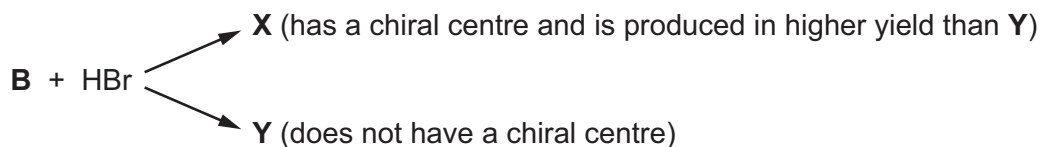
Context

- 4** **A**, **B** and **C** all have the formula C_4H_8 . They all decolourise bromine and are structural isomers of each other.

Context: part (d) (introduction)

(d) **B** does **not** show geometrical isomerism.

B reacts with HBr to form a mixture of two structural isomers, **X** and **Y**.



- (iv) Explain why **X** is produced in higher yield than **Y**.

[2]

114. M/J 17/P23/Q4(e)(ii)

Context

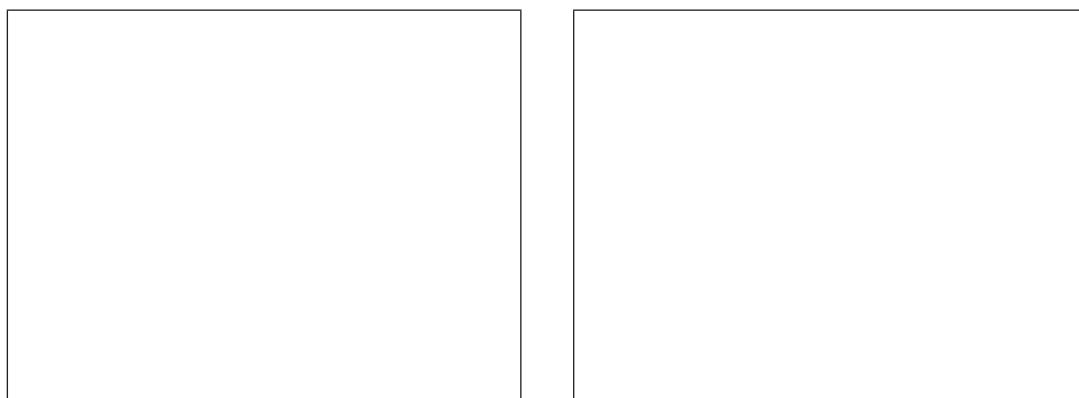
- 4** **A**, **B** and **C** all have the formula C_4H_8 . They all decolourise bromine and are structural isomers of each other.

Context: part (e) (introduction)

- (e)** **C** does **not** show geometrical isomerism.

C reacts with HBr to form a mixture of two structural isomers, neither of which has a chiral centre.

- (ii)** Draw the **displayed** formula of each of the structural isomers produced by the reaction of **C** with HBr.

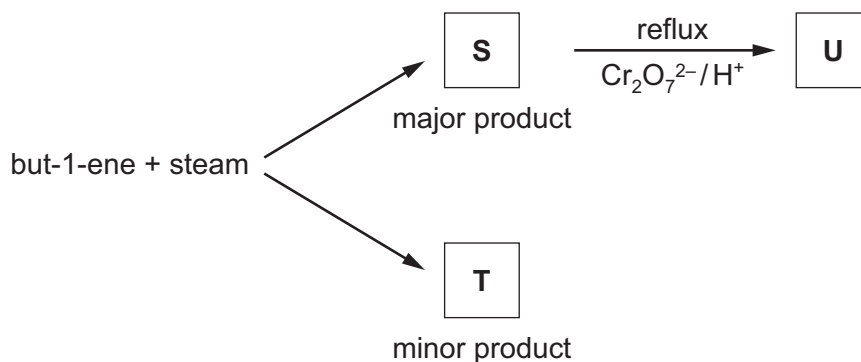


[2]

115. MAR 17/P22/Q3(c)(i)

Context: part (c) (introduction)

- (c)** But-1-ene reacts with steam as shown to form a mixture of two structural isomers, **S** and **T**.



S can be oxidised with acidified potassium dichromate(VI) to form compound **U**.
S and **U** both react with alkaline aqueous iodine.

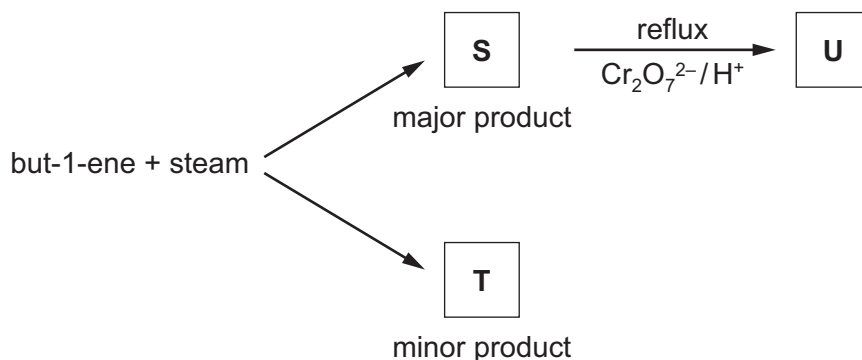
- (i)** Identify the *type of reaction* that occurs when but-1-ene reacts with steam.

[1]

116. MAR 17/P22/Q3(c)(iii)

Context: part (c) (introduction)

(c) But-1-ene reacts with steam as shown to form a mixture of two structural isomers, **S** and **T**.



S can be oxidised with acidified potassium dichromate(VI) to form compound **U**.

S and **U** both react with alkaline aqueous iodine.

(iii) Explain why **S** is the major product of the reaction of but-1-ene with steam.

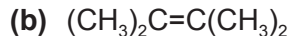
[2]

117. O/N 16/P21/Q4(b)(ii)-(iii)

Context

- 4** In each section of this question an organic compound is shown. For each compound give its name and answer the questions about it.

Context: part (b) (introduction)



- (ii) Draw the **skeletal** formula of the organic product of the reaction of this compound with cold, dilute, acidified manganate(VII) ions.

[1]

- (iii) Name the organic product of the reaction of this compound with hot, concentrated, acidified manganate(VII) ions.

[1]

118. O/N 16/P21/Q4(c)(ii)-(iii)

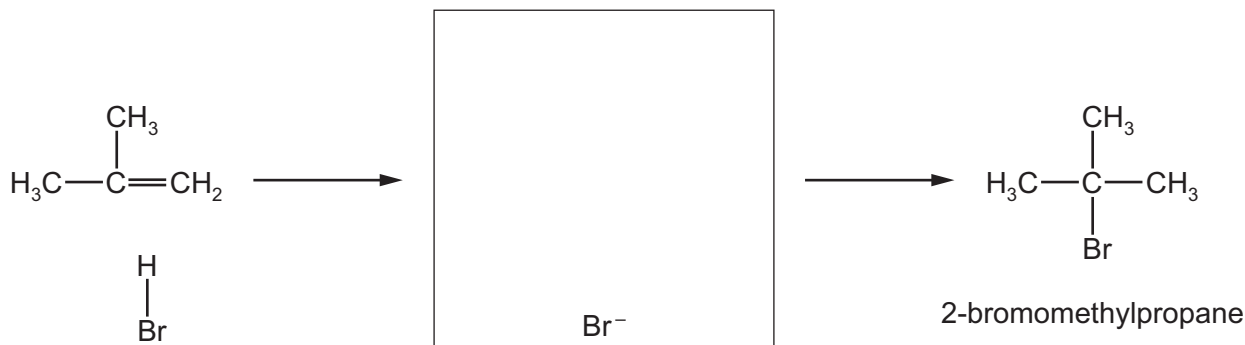
Context

- 4** In each section of this question an organic compound is shown. For each compound give its name and answer the questions about it.

Context: part (c) (introduction)

(c) $(\text{CH}_3)_2\text{C}=\text{CH}_2$

- (ii)** Complete the mechanism for the reaction of this compound with hydrogen bromide. Include all necessary curly arrows, lone pairs, charges and partial charges.



[4]

- (iii)** Explain fully why 2-bromomethylpropane is the major product of this reaction while only relatively small amounts of 1-bromomethylpropane are produced.

[3]

119. O/N 16/P22/Q5(a)(i)

Context

- 5** In each section of this question choose the answer or answers from the options listed.

Context: part (a) (introduction)

(a) Six particles are listed.



- (i)** Identify **two** particles produced during the reaction of methane and chlorine in the presence of UV light.

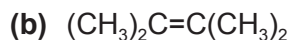
[1]

120. O/N 16/P23/Q4(b)(ii)-(iii)

Context

- 4** In each section of this question an organic compound is shown. For each compound give its name and answer the questions about it.

Context: part (b) (introduction)



- (ii)** Draw the **skeletal** formula of the organic product of the reaction of this compound with cold, dilute, acidified manganate(VII) ions.

[1]

- (iii)** Name the organic product of the reaction of this compound with hot, concentrated, acidified manganate(VII) ions.

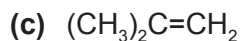
[1]

121. O/N 16/P23/Q4(c)(ii)-(iii)

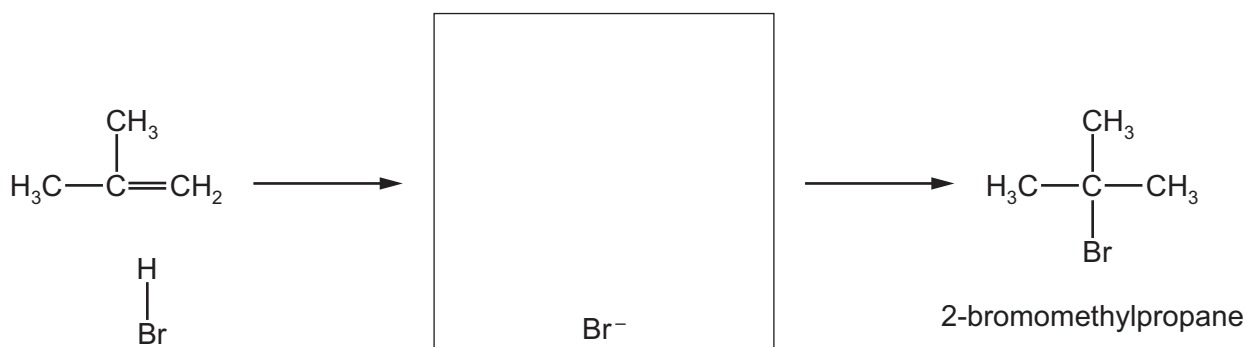
Context

- 4** In each section of this question an organic compound is shown. For each compound give its name and answer the questions about it.

Context: part (c) (introduction)



- (ii)** Complete the mechanism for the reaction of this compound with hydrogen bromide. Include all necessary curly arrows, lone pairs, charges and partial charges.



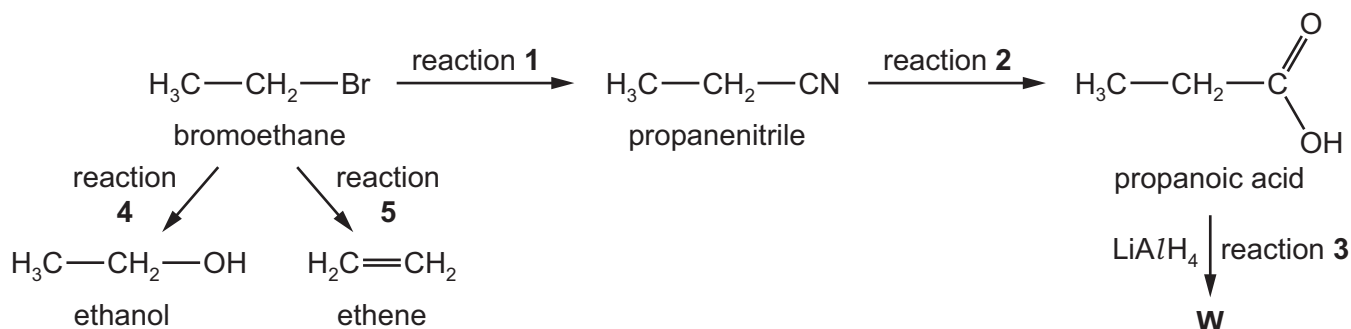
[4]

- (iii)** Explain fully why 2-bromomethylpropane is the major product of this reaction while only relatively small amounts of 1-bromomethylpropane are produced.

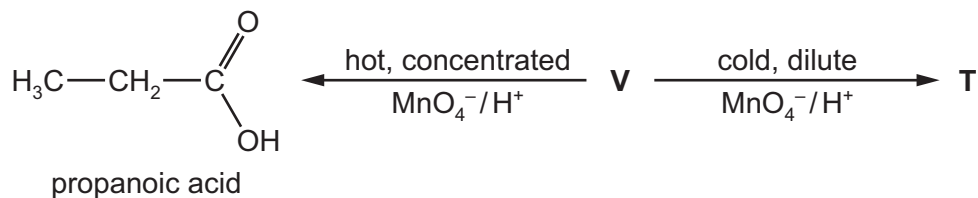
[3]

122. M/J 16/P21/Q5(e)(i)

Context

5 A reaction sequence is shown.

Context: part (e) (introduction)

(e) **V** reacts with acidified manganate(VII) ions in two different ways depending on the conditions, as shown in the reaction sequence below.**V** decolourises bromine water.

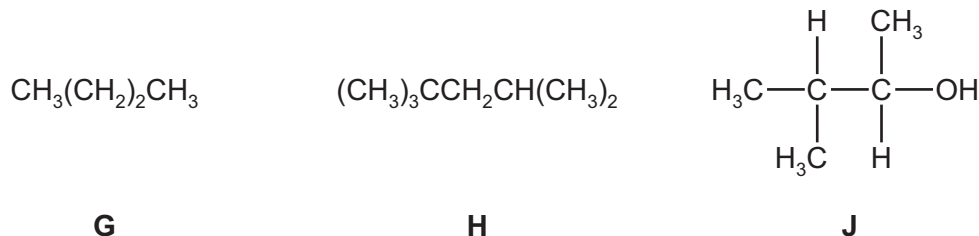
When the acidified manganate(VII) is hot and concentrated, propanoic acid is the only organic product.

When the acidified manganate(VII) is cold and dilute, the organic product is **T** which has two chiral centres.**(i)** Give the structural formulae of **V** and **T**.**V****T**

[2]

123. MAR 16/P22/Q4(b)

Context

4 The following compounds were all found to be components of a sample of petrol.**(b)** Write an equation to represent the complete combustion of compound **H**.

[1]

124. O/N 15/P21/Q3(b)

Context

- 3** Heptane, C_7H_{16} , is an undesirable component of petrol as it burns explosively causing 'knocking' in an engine.

(b) (i) Write an equation for the complete combustion of heptane.

[1]

(ii) Write an equation for the incomplete combustion of heptane leading to the production of a solid pollutant.

[1]

(iii) Incomplete combustion can also lead to emission of unburnt hydrocarbons.

State one environmental consequence of this.

[1]

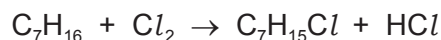
125. O/N 15/P21/Q3(c)

Context

- 3** Heptane, C_7H_{16} , is an undesirable component of petrol as it burns explosively causing 'knocking' in an engine.

(c) The reaction of heptane with chlorine in the presence of UV light produces a wide variety of products.

Formation of the monochloroheptanes can be represented by the following equation.



(i) Name the mechanism of the reaction between heptane and chlorine in the presence of UV light.

[1]

(ii) Describe this mechanism, using suitable equations and including the names of each stage in the process.

[5]

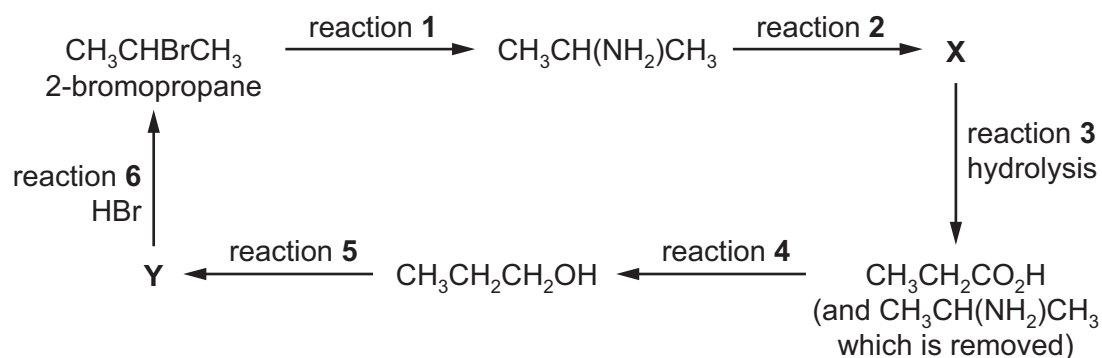
126. O/N 15/P22/Q4(b)(iv)

Context

4 Halogenoalkanes are useful intermediates in the synthesis of a wide variety of compounds.

Context: part (b) (introduction)

(b) Some reactions involving 2-bromopropane are shown.



(iv) Name the type of reaction involved in reaction **4** and suggest a suitable reagent.

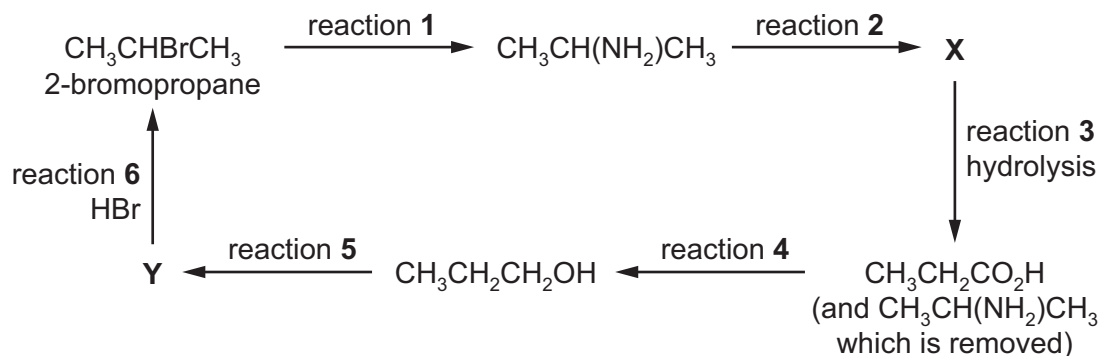
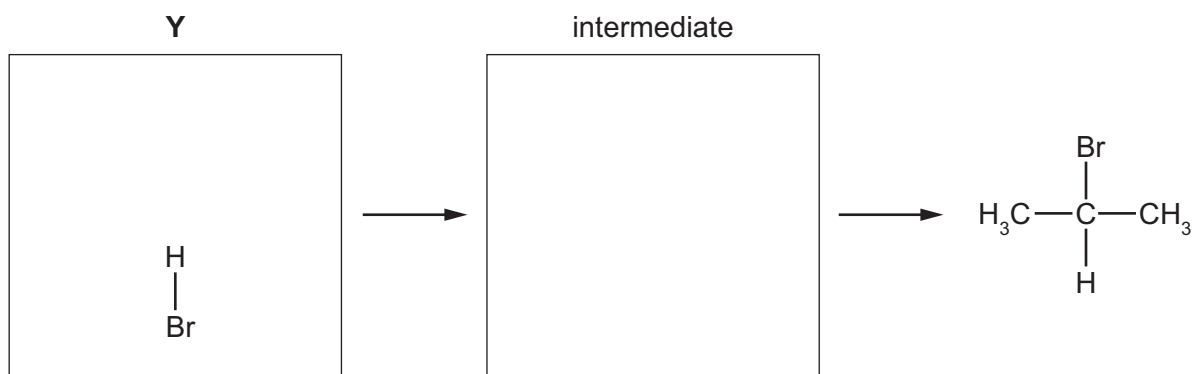
[2]

127. O/N 15/P22/Q4(b)(vi)-(viii)

Context

4 Halogenoalkanes are useful intermediates in the synthesis of a wide variety of compounds.

Context: part (b) (introduction)

(b) Some reactions involving 2-bromopropane are shown.**(vi)** Complete the mechanism for the production of 2-bromopropane from **Y** in reaction 6 shown below.Include the structure of **Y** and any necessary lone pairs, curly arrows, charges and partial charges.

[4]

(vii) Give the name of the mechanism in **(vi)**.

[1]

(viii) 1-bromopropane is a minor product of reaction 6.

Explain why 2-bromopropane is the major product of reaction 6.

[2]

128. O/N 15/P23/Q3(b)

Context

- 3** Heptane, C_7H_{16} , is an undesirable component of petrol as it burns explosively causing 'knocking' in an engine.

(b) (i) Write an equation for the complete combustion of heptane.

[1]

(ii) Write an equation for the incomplete combustion of heptane leading to the production of a solid pollutant.

[1]

(iii) Incomplete combustion can also lead to emission of unburnt hydrocarbons.

State one environmental consequence of this.

[1]

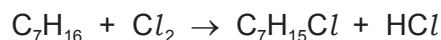
129. O/N 15/P23/Q3(c)

Context

- 3** Heptane, C_7H_{16} , is an undesirable component of petrol as it burns explosively causing 'knocking' in an engine.

(c) The reaction of heptane with chlorine in the presence of UV light produces a wide variety of products.

Formation of the monochloroheptanes can be represented by the following equation.



(i) Name the mechanism of the reaction between heptane and chlorine in the presence of UV light.

[1]

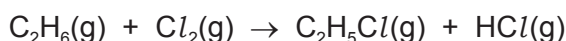
(ii) Describe this mechanism, using suitable equations and including the names of each stage in the process.

[5]

130. M/J 15/P22/Q3(b)(iii)

Context

- 3** Ethane reacts with chlorine to form chloroethane.



Context: part (b) (introduction)

(b) Chloroethane can be converted back into ethane by a two-stage process via an intermediate compound, **X**.

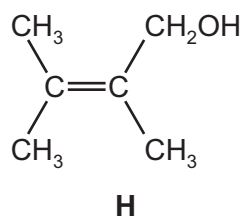


(iii) Suggest the reagent and conditions needed for reaction 2.

[1]

131. M/J 15/P23/Q4(a)

Context

4 The structure of **H** is shown.

(a) **H** reacts with both cold, dilute, acidified potassium manganate(VII) and with hot, concentrated, acidified potassium manganate(VII).

(i) Give the structure of the organic product of the reaction of **H** with cold, dilute, acidified potassium manganate(VII).

[1]

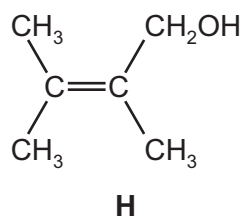
(ii) Give the structures of the organic products of the reaction of **H** with hot, concentrated, acidified potassium manganate(VII).

[2]

132. M/J 15/P23/Q4(b)

also Unit 13 (3 marks); this unit 4 marks

Context

4 The structure of **H** is shown.

- (b) (i)** Complete the reaction scheme to show the mechanism of the reaction of **H** with bromine to form **J**.

Include all necessary curly arrows, lone pairs and charges.



[3]

- (ii)** Explain the origin of the dipole on the bromine molecule.

[1]

J is formed as an equimolar mixture of isomers.

- (iii)** State the type of isomerism shown by **J**.

[1]

- (iv)** Draw the structures of the two isomers of **J**.

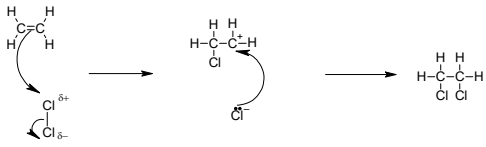
[2]

Answers Section

1. M/J 26/P21/Q5(b)

5(b)	electrophilic addition	1	1 addition alone
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2. M/J 26/P21/Q5(d)

5(d)	 M1 curly arrow from C=C bond to Cl closest to C=C M2 correct dipoles $Cl\delta^+ - Cl\delta^-$ AND curly arrow from bond of $Cl\delta^+ - Cl\delta^-$ to $Cl\delta^-$ M3 correct carbocation intermediate M4 curly arrow from lone pair of $:Cl^-$ to C^+ of intermediate Judgement of arrows: A curly arrows that appear straight NOT single headed arrows (fishhooks) first time seen-and then A M1 judgement of arrows: π bonds M1 starts from anywhere on π bond AND goes to Cl or close to Cl atom of $Cl-Cl$ M1 if arrow does not start at the double bond, but when it is extrapolated back the base of the arrow hits the π bond AND the arrow end goes to Cl or close to Cl atom of $Cl-Cl$ M2 judgement of arrows: $Cl-Cl$ bond M2 A curly arrow from bond of $Cl-Cl$ to just beyond the Cl (if dipole correct).	4	M1 δ^-. I π or δ^- label on the double bond (e.g. $C=C$) I partial charges on Cl R partial charges on C of $C=C$ (e.g. $\delta^+C = \delta^+C / \delta^-C = \delta^-C$) R arrows from any other part of the alkene M2 A dipole moment of Cl_2 shown as an arrow with + convention I electrons (i.e. dots / crosses above and/or below bond) M3 A unambiguous structure R δ^+ charge on C^+ R lone pair on C^+ M4 A ecf from incorrect intermediate structure in M3 when no positive charge on C of intermediate OR $C\delta^+$ instead of C^+ I δ^-Cl^- if all else ok
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3. M/J 26/P22/Q5(b)(iii)-(iv)

5(b)(iii)	M1 $CH_3(CH_2)_4CH_3 + Cl\cdot \rightarrow CH_3(CH_2)_3\cdot CHCH_3 + HCl$ M2 $CH_3(CH_2)_3\cdot CHCH_3 + Cl_2 \rightarrow CH_3(CH_2)_3CHClCH_3 + Cl\cdot$	2	Independent marks A radicals with or without $\cdot$ M1 I alkyl radical as $\cdot C_6H_{13}$ OR $CH_3(CH_2)_5\cdot$ I position of the dot BOD $CH_3(CH_2)_3CH_3CH\cdot$ OR $CH_3(CH_2)_3CCH_3H\cdot$ M2 A ecf on their alkyl structure from M1 (even if M2 does not balance) M2 A alkyl radical as $\cdot C_6H_{13}$ OR $CH_3(CH_2)_5\cdot$ M2 ecf use of anion/cation
5(b)(iv)	M1 $CH_3(CH_2)_4CH_3 + \cdot Cl \rightarrow CH_3(CH_2)_4\cdot CH_2 + HCl$ M2 $CH_3(CH_2)_4\cdot CH_2 + CH_3(CH_2)_4\cdot CH_2 \rightarrow CH_3(CH_2)_{10}CH_3$ OR $2CH_3(CH_2)_4\cdot CH_2 \rightarrow CH_3(CH_2)_{10}CH_3$	2	Independent marks A radicals with or without $\cdot$ M1 A $CH_3(CH_2)_5\cdot$ M1 I alkyl radical as $\cdot C_6H_{13}$ I position of the dot M2 A alkyl radical as $\cdot C_6H_{13}$ OR $CH_3(CH_2)_5\cdot$ M2 ecf use of anion/cation

4. M/J 26/P22/Q5(c)

5(c)	number of alkyl / methyl groups	1	A number of C atoms bonded to the central carbon / branches / R groups BOD number of alkyl groups bonded to the carbocation I number of C atoms bonded to the carbocation I number of C atoms I number of ethyl/propyl groups I number of hydrogens I inductive effect I tertiary>secondary>primary stability I carbocation stability
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5. M/J 26/P22/Q6(b)(i)

6(b)(i)	M1 steam / H ₂ O (g) M2 H ₃ PO ₄ (catalyst) / phosphoric (V) acid (catalyst)	2	M1 A water <u>and</u> then temperature $\geq 100^\circ\text{C}$ (temperature might be seen on conditions line) M2 A H ₃ PO ₄ (catalyst) / phosphoric (V) acid (catalyst) / sulfuric acid (catalyst) / H ₂ SO ₄ (catalyst) on reagent line A orthophosphoric acid / phosphoric acid / sulfuric acid I heat under reflux / heat alone I pressure I state symbols
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6. M/J 26/P23/Q5(a)(ii)

5(a)(ii)	free-radical substitution	1	
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7. M/J 26/P23/Q5(b)(ii)-(iii)

5(b)(ii)	M1 $\text{CH}_3\text{CH}_2\text{CH}_3 + \bullet\text{Cl} \rightarrow \bullet\text{CH}_2\text{CH}_2\text{CH}_3 + \text{HCl}$ M2 $\bullet\text{CH}_2\text{CH}_2\text{CH}_3 + \text{Cl}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \bullet\text{Cl}$	2	M1 I $\bullet\text{C}_3\text{H}_7$ M1 R $\text{CH}_3\bullet\text{CHCH}_3$ I position of the dot I absence of the dot M2 A alkyl radical as $\bullet\text{C}_3\text{H}_7$ or $\text{CH}_3\bullet\text{CHCH}_3$
5(b)(iii)	M1 $\text{CH}_3\text{CH}_2\text{CH}_3 + \bullet\text{Cl} \rightarrow \text{CH}_3\bullet\text{CHCH}_3 + \text{HCl}$ M2 $2\text{CH}_3\bullet\text{CHCH}_3 \rightarrow (\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2	A radicals with or without $\bullet$ $(\text{CH}_3)_2\text{CH}\bullet = \text{CH}_3\bullet\text{CHCH}_3$ M1 A C_3H_7 for CH_3CHCH_3 M1 R $\bullet\text{CH}_2\text{CH}_2\text{CH}_3$ I position of the dot M2 I alkyl radical as $\bullet\text{C}_3\text{H}_7$ M2 R $\bullet\text{CH}_2\text{CH}_2\text{CH}_3$

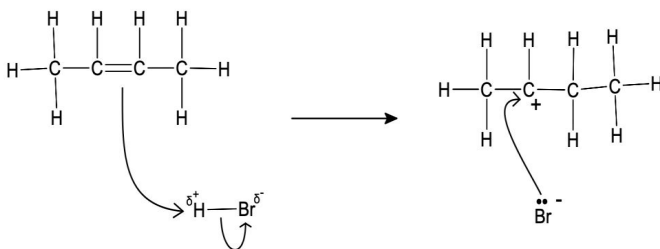
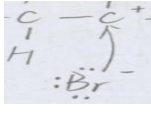
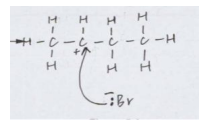
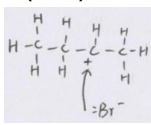
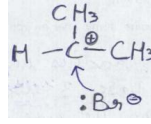
8. M/J 26/P24/Q5(a)

5(a)	Al ₂ O ₃ AND heat / high temperature	1	A zeolite / alumina / porous pot / porcelain / ceramics AND heat A high temperature/high temp = heat A $\geq 100^\circ\text{C}$ I catalyst I steam I pressure
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9. M/J 26/P24/Q5(b)(iii)

5(b)(iii)	orange / brown (solution / liquid) to colourless	1	I turns clear I colour disappears I discolouration / colour discharged A orange / brown disappears A decolourises / turns colourless A brown / yellow / yellow-brown / orange-brown / yellow-orange R red / red-brown I 'liquid'
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10. M/J 26/P24/Q5(d)

5(d)(i)	electron pair acceptor	1	I reference to likes electrons / attracted to electrons / area of low electron density. A lone pair acceptor A gains both electrons I electron acceptor
5(d)(ii)	 M1 curly arrow from $=$ of $C=C$ to H - I π or $\delta-$ label on the double bond - R partial charges shown on C of $C=C$ R arrows from any other part of the alkene M2 correct dipoles $H^{\delta+}-Br^{\delta-}$ AND curly arrow from bond of $H^{\delta+}-Br^{\delta-}$ to $Br^{\delta-}$ - I electrons (i.e. dots / crosses above and / or below bond) M3 correct intermediate – secondary carbocation only - A unambiguous structure - I $\delta+$ charge on C^+ - R lone pair on C^+	4	Judgement of arrows: A curly arrows that appear straight R single headed arrows first time seen and then A M1 judgement of arrows: π bonds M1 starts from anywhere on π bond AND goes to H or close to H atom of $H-Br$ ✓ M1 if arrow does not start at the double bond, but when it is extrapolated back the base of the arrow hits the π bond AND the arrow end goes to H or close to H atom of $H-Br$ ✓ M2 judgement of arrows: $H-Br$ bond M2 A curly arrow from bond of $H-Br$ to just beyond the Br (if dipole correct).
5(d)(ii) ctd.	M4 curly arrow from lone pair of $:Br^-$ to C^+ of intermediate A ecf from incorrect intermediate structure in M3 when: - no positive charge on C of intermediate OR $C^{\delta+}$ instead of C^+ - I $\delta-$ on $:Br^-$		Judgement of arrows: lone pairs M4 A arrow starts anywhere closely associated with lone pair A  A  A (BOD)  R 

11. MAR 26/P22/Q3(c)

3(c)(i)	(free-)radical substitution	1
3(c)(ii)	M1 <i>propagation</i> $\text{Cl}^\bullet + \text{C}_4\text{H}_{10} \rightarrow \text{C}_4\text{H}_9^\bullet + \text{HCl}$ M2 <i>termination</i> $\text{C}_4\text{H}_9^\bullet + \text{Cl}^\bullet \rightarrow \text{C}_4\text{H}_9\text{Cl}$	2

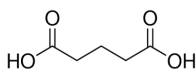
12. O/N 25/P21/Q3(c)(ii)-(iii)

3(c)(ii)	M1 $\text{C}_5\text{H}_{10} + \text{Cl}^\bullet \rightarrow \text{C}_5\text{H}_9^\bullet + \text{HCl}$ M2 $\text{C}_5\text{H}_9^\bullet + \text{Cl}_2 \rightarrow \text{C}_5\text{H}_9\text{Cl} + \text{Cl}^\bullet$	2
3(c)(iii)	termination	1

13. O/N 25/P21/Q3(d)(i)

3(d)(i)	HCl (g)	1
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14. O/N 25/P21/Q3(e)(i)

3(e)(i)		1
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15. O/N 25/P22/Q3(b)

3(b)	M1 strong C—H (bonds) M2 non-polar (C—H / molecule / bonds)	2
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16. O/N 25/P22/Q3(c)

3(c)	cracking	1
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17. O/N 25/P22/Q3(e)(i)

3(e)(i)	H_2 ((g)) OR hydrogen (gas)	1
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18. O/N 25/P22/Q3(e)(iii)

3(e)(iii)	$\text{C}_6\text{H}_{14} + 9\frac{1}{2}\text{O}_2 \rightarrow 6\text{CO}_2 + 7\text{H}_2\text{O}$	1
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19. O/N 25/P22/Q4(a)(i)-(ii)

4(a)(i)	free-radical substitution	1
4(a)(ii)	M1 $\text{C}_3\text{H}_8 + \text{Cl}^\bullet \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2^\bullet + \text{HCl}$ OR $\text{C}_3\text{H}_8 + \text{Cl}^\bullet \rightarrow \text{C}_3\text{H}_7^\bullet + \text{HCl}$ M2 $\text{CH}_3\text{CH}_2\text{CH}_2^\bullet + \text{Cl}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \text{Cl}^\bullet$ OR $\text{C}_3\text{H}_7^\bullet + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_7\text{Cl} + \text{Cl}^\bullet$ etc M3 $\text{CH}_3\text{CH}_2\text{CH}_2^\bullet + \text{Cl}^\bullet \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$ OR $\text{C}_3\text{H}_7^\bullet + \text{Cl}^\bullet \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$	3

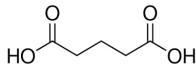
20. O/N 25/P23/Q3(c)(ii)-(iii)

3(c)(ii)	M1 $\text{C}_5\text{H}_{10} + \text{Cl}^\bullet \rightarrow \text{C}_5\text{H}_9^\bullet + \text{HCl}$ M2 $\text{C}_5\text{H}_9^\bullet + \text{Cl}_2 \rightarrow \text{C}_5\text{H}_9\text{Cl} + \text{Cl}^\bullet$	2
3(c)(iii)	termination	1

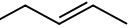
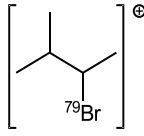
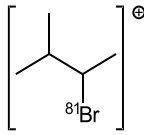
21. O/N 25/P23/Q3(d)(i)

3(d)(i)	HCl (g)	1
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22. O/N 25/P23/Q3(e)(i)

3(e)(i)		1
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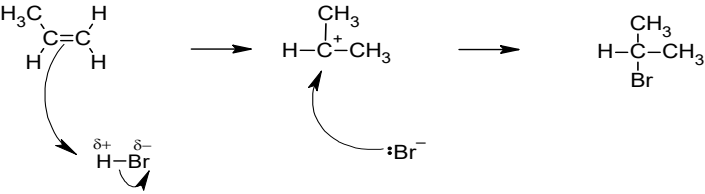
23. O/N 25/P24/Q4(a)

4(a)(i)	M1 (molecules with the) same molecular formula / same number of atoms of each element M2 different structural formulae/structures	2
4(a)(ii)	M1  M2 restricted / lack of / no rotation about C=C	2
4(a)(iii)	electrophilic addition	1
4(a)(iv)	M1 CH ₃ COCH ₃ + CH ₃ COOH as products M2 (CH ₃) ₂ C = CHCH ₃ + 3[O] → CH ₃ COCH ₃ + CH ₃ COOH	2
4(a)(v)	M1 (reaction proceeds via a) more stable intermediate / carbocation M2 because of greater positive inductive effect of more alkyl groups	2
4(a)(vi)	M1  M2 	2

24. M/J 25/P21/Q4(b)(iv)

4(b)(iv)	M1 (also) oxidises / reacts with (primary and / or secondary) alcohol groups M2 (also) breaks the sigma and pi bond in C=C	2
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25. M/J 25/P22/Q3(a)(ii)-(iii)

3(a)(ii)	 M1 curly arrow from = of C=C to H (of HBr) M2 curly arrow from bond of H ^{δ+} —Br ^{δ-} to Br ^{δ-} (dipole is essential) M3 correct intermediate M4 curly arrow from lone pair of Br ⁻ to C ⁺ of intermediate drawn	4
3(a)(iii)	M1 (2-bromopropane is made from the) more stable intermediate / carbocation M2 (due to positive) inductive effect of more alkyl groups (attached to the C ⁺ / central carbon atom) OR (due to) greater (positive) inductive effect of 2 methyl groups (rather than 1 ethyl group)	2

26. M/J 25/P22/Q3(e)

3(e)	$\text{H}_2\text{C}=\text{CHCH}_2\text{CH}=\text{C}(\text{CH}_3)_2$ ✓✓  ✓✓ $\text{H}_2\text{CCHCH}_2\text{CHC}(\text{CH}_3)_2$ ✓✓ M1 $\text{H}_2\text{C}=\text{C}\dots$ on one end of their molecule OR $\dots\text{C}=\text{C}(\text{CH}_3)_2$ on one end of their molecule OR $\dots\text{C}=\text{CHCH}_2\text{CH}=\text{C}\dots$ in the middle of their molecule M2 correct structure	2
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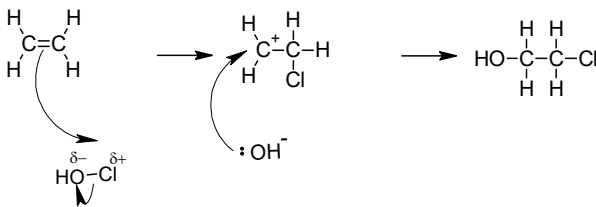
27. M/J 25/P23/Q3(a)(i)

3(a)(i)	Al_2O_3 / aluminium oxide	1
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28. M/J 25/P23/Q3(b)(i)

3(b)(i)	nickel / platinum	1
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29. M/J 25/P23/Q6(b)

6(b)	 M1 correct arrow for HO-Cl M2 arrow from double bond of C=C to Cl M3 correct intermediate M4 arrow from lone pair on O of :OH- to the appropriate C	4
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30. M/J 25/P24/Q5(d)

5(d)(i)	oxidising agent	1
5(d)(ii)	hot concentrated acidified KMnO_4	1

31. MAR 25/P22/Q3(d)

3(d)(i)	instantaneous dipole—induced dipole / id—id / London / dispersion (forces) AND permanent dipole—permanent dipole / pd—pd (forces)	1
3(d)(ii)	electrophilic addition	1
3(d)(iii)		2
3(d)(iv)	position(al) / regio- (isomerism)	1
3(d)(v)	- secondary (carbo)(cat)ion/intermediate is more stable / $[\text{CH}_3\text{CHCH}_2\text{I}]^+$ is preferred - electrophile approaches with I nearest / $\text{I}^{\delta+}$ / I^+ attacks first - I and Br have different electronegativities / IBr is a polar molecule	2

32. O/N 24/P21/Q2(b)(ii)-(iii)

2(b)(ii)	ultraviolet	1
2(b)(iii)	$\text{C}_2\text{H}_6 + \text{Cl}^\bullet \rightarrow \text{C}_2\text{H}_5^\bullet + \text{HCl}$	1
	$\text{C}_2\text{H}_5^\bullet + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{Cl}^\bullet$	1

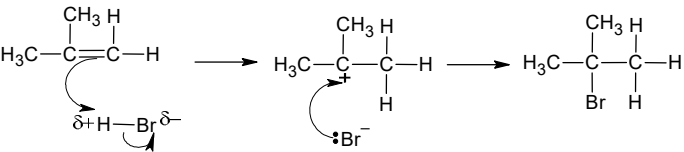
33. O/N 24/P22/Q2(b)(iii)

2(b)(iii)		2
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34. O/N 24/P23/Q2(b)(ii)-(iii)

2(b)(ii)	ultraviolet	1
2(b)(iii)	$C_2H_6 + Cl^\bullet \rightarrow C_2H_5^\bullet + HCl$	1
	$C_2H_5^\bullet + Cl_2 \rightarrow C_2H_5Cl + Cl^\bullet$	1

35. M/J 24/P21/Q5(c)(ii)

5(c)(ii)	 M1 arrow from = of C=C to H of H-Br M2 correct dipole on H-Br AND arrow from H-Br bond to Br M3 correct intermediate AND 2-bromopropane M4 arrow from lone pair on Br⁻ to C⁺ of their intermediate (with +ve or δ+ charge)	4
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36. M/J 24/P21/Q6(b)(ii)

6(b)(ii)	sodium borohydride / NaBH ₄	1
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37. M/J 24/P23/Q5(a)

5(a)	cold dilute KMnO ₄ OR cold dilute potassium manganate(VII)	1
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38. M/J 24/P23/Q5(b)(ii)

5(b)(ii)	electrophilic addition	1
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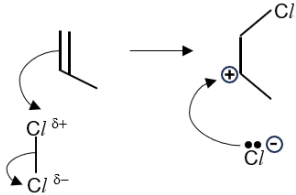
Answer for context part (a)

5(a)	cold dilute KMnO ₄ OR cold dilute potassium manganate(VII)	1
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39. MAR 24/P22/Q1(d)(ii)

1(d)(ii)	(KMnO ₄ is) cold and dilute	1
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40. MAR 24/P22/Q2(d)(i)

2(d)(i)	 • curly arrow from C=C bond to Cl(δ⁺) • correct dipole on Cl₂ AND curly arrow from Cl-Cl bond to Cl^{δ-} • correct carbocation intermediate • curly arrow from lone pair on Cl⁻ to C⁺	1
	• curly arrow from C=C bond to Cl(δ ⁺)	1
	• correct dipole on Cl ₂ AND curly arrow from Cl-Cl bond to Cl ^{δ-}	1
	• correct carbocation intermediate	1
	• curly arrow from lone pair on Cl ⁻ to C ⁺	1

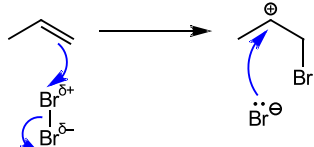
41. MAR 24/P22/Q4(b)(i)

4(b)(i)	molecule	P	Q	R	S	1
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42. O/N 23/P22/Q3(c)(i)

3(c)(i)	H ₂ O / steam	1
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43. O/N 23/P22/Q4(b)(i)

4(b)(i)	 curly arrow from C=C bond to Br^{δ+} AND curly arrow from Br^{δ+}—Br^{δ-} bond to Br^{δ-} AND Br^{δ+} closest to the double bond correct intermediate curly arrow from lone pair on Br⁻ to C⁽⁺⁾ of the intermediate	1
		1
		1

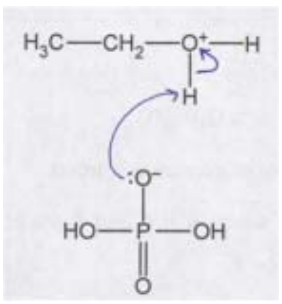
44. M/J 23/P21/Q3(d)

3(d)(i)	ultraviolet light			1	
3(d)(ii)		name of stage	equation		
		initiation	$\text{Cl}_2 \rightarrow 2\text{Cl}\cdot$		[1]
		propagation	$\text{CH}_2\text{Cl}_2 + \text{Cl}\cdot \rightarrow \text{CHCl}_2\cdot + \text{HCl}$		[1]
		termination	$\text{CHCl}_2\cdot + \text{Cl}\cdot \rightarrow \text{CHCl}_3$		[1]

45. M/J 23/P21/Q5(a)

5(a)(i)	$\text{C}_8\text{H}_{18} + 12\frac{1}{2} \text{O}_2 \rightarrow 8\text{CO}_2 + 9\text{H}_2\text{O}$ M1 correct formula of M C ₈ H ₁₈ M2 balanced equation showing complete combustion of M	2
5(a)(ii)	• C • CO • unburnt hydrocarbons • less/no CO₂ Two correct for one mark, three correct for two marks.	2

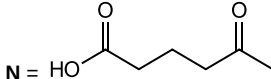
46. M/J 23/P22/Q4(c)(ii)-(iii)

4(c)(ii)	(CH ₃ CH ₂) ⁺ OR CH ₃ C ⁺ H ₂ AND accepts a pair of electrons	1
4(c)(iii)	 M1 arrow from bond of one of the O-H to O of the cation M2 arrow from lone pair on O of the anion to between H of H-O	2

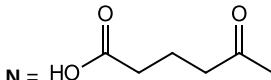
47. M/J 23/P22/Q4(c)(vi)

4(c)(vi)	M1 (propan-2-ol is produced via a) more stable carbocation OR more stable intermediate OR $(\text{CH}_3)_2\text{C}^+\text{H}$ is more stable M2 (positive) inductive effect of more alkyl/methyl groups OR more electron donating alkyl / methyl groups OR greater electron donating effect of two alkyl / two methyl groups	2
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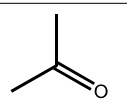
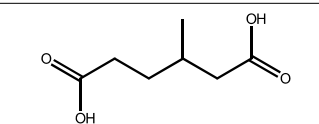
48. O/N 22/P21/Q5(a)(i)

5(a)(i)		1
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49. O/N 22/P23/Q5(a)(i)

5(a)(i)		1
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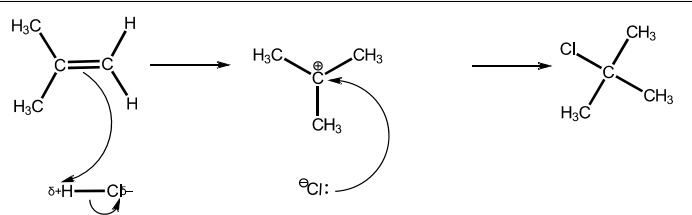
50. M/J 22/P21/Q3(b)

3(b)	 OR $(\text{CH}_3)_2\text{CO}$  OR $\text{HOOC}(\text{CH}_2)_2\text{CH}(\text{CH}_3)\text{CH}_2\text{COOH}$	2
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51. M/J 22/P21/Q3(c)(ii)-(iii)

3(c)(ii)	M1 $\text{H}_2\text{O}(\text{g})$ / steam M2 phosphoric acid (catalyst) / H_3PO_4 (catalyst)	2
3(c)(iii)	M1 Q AND intermediate / cation is (more) stable M2 (as the) C^+ has 3 / more alkyl groups attached M3 (so) greater (positive) inductive effect	3

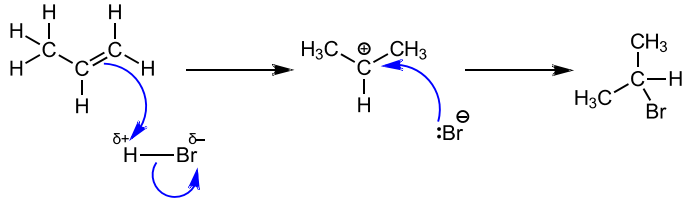
52. M/J 22/P23/Q4(a)

4(a)(i)	 M1 curly arrow from = to H AND correct dipole on H-Cl M2 correct intermediate M3 curly arrow from H-Cl bond to Cl AND curly arrow from lp on: Cl^- to C^+	3
4(a)(ii)	- intermediate OR C^+ OR carbocation (formed) is more stable - due to greater (positive) inductive effect. - caused by 3 methyl groups / 3 alkyl groups / more alkyl groups attached to C^+ / central C	2

53. M/J 22/P23/Q5(a)(i)

5(a)(i)	cold + dilute	1
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54. MAR 22/P22/Q3(e)(i)-(ii)

3(e)(i)	 M1: curly arrow from C=C to H of HBr M2: correct dipole ($\delta^+ \text{H}-\text{Br}^{\delta-}$) AND curly arrow from H—Br to Br M3: curly arrow from lone pair on $:\text{Br}^-$ to carbocation M4: correct intermediate AND product (2-bromopropane)	4
3(e)(ii)	2° carbocation more stable (than 1°) (or reverse argument) [1] greater positive inductive effect of two methyl groups (<i>cf.</i> one ethyl) (or reverse argument) [1]	2

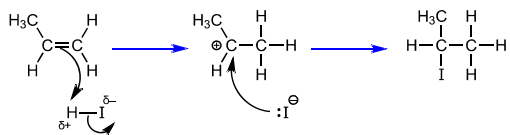
55. O/N 21/P21/Q3(d)

3(d)	CH₃CH₂CO₂H CH₃COCH₃	2
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56. O/N 21/P21/Q4(a)(iii)

4(a)(iii)	substitution	1
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57. O/N 21/P22/Q1(g)

1(g)(i)	M1 2-iodopropane – formed from a (more) stable (secondary) (carbo)cation/intermediate M2 (because of) greater (positive) inductive effect / (+)I of two alkyl groups OR (because of positive) inductive effect / (+)I of more R / more methyl / more alkyl groups	2
1(g)(ii)	 M1 curly arrow from = of C=C to H AND curly arrow from bond of H—I to I M2 curly arrow from lone pair of I⁻ to C⁽⁺⁾ of their intermediate M3 correct carbocation AND product for 2-iodopropane	3

58. O/N 21/P22/Q3(d)(i)

3(d)(i)	cold dilute acidified potassium manganate(VII)	1
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59. O/N 21/P23/Q3(d)

3(d)	CH₃CH₂CO₂H CH₃COCH₃	2
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60. O/N 21/P23/Q4(a)(iii)

4(a)(iii)	substitution	1
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61. M/J 21/P21/Q5(b)

5(b)(i)	(thermal) cracking	1
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5(b)(ii)	structure of W  OR $\text{CH}_3(\text{CH}_2)_3\text{CH}_3$ OR 	1
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62. M/J 21/P21/Q5(c)(ii)

5(c)(ii)	M1 (add) $\text{Br}_2(\text{aq})$ / bromine water	1
	M2 (solution) turns (from brown / orange / red to) colourless / decolorises OR brown / orange / red fades	1

63. M/J 21/P21/Q6(a)

6(a)	addition	1
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64. M/J 21/P21/Q6(b)

6(b)	M1 <i>catalyst</i> = sulfuric acid / phosphoric(V) acid	1
	M2 <i>conditions of reaction</i> = steam / heat (and pressure)	1

65. M/J 21/P21/Q6(d)

6(d)(i)	M1 more stable = $\text{CH}_3\text{C}^+(\text{H})(\text{CH}_3)$ 	1
	M2 less stable = $\text{CH}_3\text{CH}_2\text{C}^+(\text{H}_2)$ / 	1
	M3 greater (positive) inductive effect of two alkyl groups OR greater electron donation of two alkyl groups	1
6(d)(ii)	propan-2-ol	1

66. M/J 21/P22/Q4(b)

4(b)(i)	A	1
4(b)(ii)	M1  skeletal formula only	1
	M2 <i>explanation in terms of increased / greater stability of intermediate / carbocation</i> intermediate / (secondary) carbocation / $\text{CH}_3\text{C}^+(\text{H})(\text{CH}_2\text{CH}_3)$ is (more) stable	1
	M3 <i>reason for increased stability of intermediate in terms of greater number of alkyl groups showing largest inductive (electron releasing) effect</i> greater (positive) inductive effect due to two alkyl groups OR greater electron donation of two alkyl groups	1

67. MAR 21/P22/Q2(g)(i)-(ii)

2(g)(i)	(free-)radical substitution	1
2(g)(ii)	ultraviolet (UV) light / sunlight	1

68. MAR 21/P22/Q2(g)(iv)-(vi)

2(g)(iv)	$\text{Cl}^\cdot$ AND CH_2Cl_2	1
2(g)(v)	termination	1
2(g)(vi)	CHCl_3 OR $(\text{CH}_2\text{Cl})_2$	1

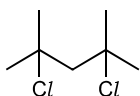
69. MAR 21/P22/Q4(d)(iii)

4(d)(iii)	ethene	1
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70. O/N 20/P21/Q4(b)(ii)

4(b)(ii)	Electrophilic addition	1
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71. O/N 20/P22/Q3(c)(iii)

3(c)(iii)		1
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72. O/N 20/P23/Q4(b)(ii)

4(b)(ii)	Electrophilic addition	1
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73. M/J 20/P21/Q5(a)(i)-(ii)

5(a)(i)	Cl	1
5(a)(ii)	HCl AND H ₂ O	1

74. M/J 20/P21/Q5(b)(i)

5(b)(i)	ultraviolet light / uv	1
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75. M/J 20/P21/Q5(d)

5(d)(i)	<i>Two structures representing the intermediate</i> M1 C ₂ H ₅ C ⁺ HCH ₃ M2 CH ₃ CH ₂ CH ₂ C ⁺ H ₂	2
5(d)(ii)	<i>Identify the most stable intermediate</i> M1 C ₂ H ₅ C ⁺ HCH ₃ <i>explanation</i> M2 (more / 2 alkyl groups attached so) it has the greater inductive / electron donating effect	2

76. M/J 20/P21/Q6(b)

6(b)(i)	hot AND concentrated	1
6(b)(ii)	oxidation	1

77. M/J 20/P21/Q6(c)

6(c)	<i>Structural formula of X:</i> HCO ₂ H OR HCOOH	1
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78. MAR 20/P22/Q3(c)(iii)

3(c)(iii)	M1 curly arrow from C=C double bond to Br M2 correct dipole in Br₂ AND curly arrow from Br—Br to Br^{δ-} M3 correct intermediate AND curly arrow from lone pair on Br⁻ to C⁺ M4 correct product	4
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Answer for context part (a)(v)

3(a)(v)	M1 hydrolysis M2 esterification / condensation	2
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79. MAR 20/P22/Q3(d)(ii)

3(d)(ii)	H ₂ / hydrogen	1
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80. O/N 19/P21/Q1(e)

1(e)(i)	ultra-violet (light) / sunlight	1
1(e)(ii)	$\text{C}_2\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{HCl}$	1

81. O/N 19/P21/Q3(a)(i)

3(a)(i)	cracking	1
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82. O/N 19/P21/Q3(a)(iii)

3(a)(iii)	$\text{C}_4\text{H}_8 + 4\text{O}_2 \rightarrow 4\text{CO} + 4\text{H}_2\text{O}$	1
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83. O/N 19/P22/Q4(c)(i)

4(c)(i)	Add bromine water / Br ₂ (aq) AND turns (from orange / brown to) colourless	1
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84. O/N 19/P22/Q4(e)(iv)

4(e)(iv)	M1: recognise this reaction involves less stable intermediate 1° (carbo)cation (intermediate) is less stable (than 3°) M2: explain difference in reactivity in terms of positive inductive effect – comparative answer lower (positive) inductive effect / lower (+)I OR inductive effect of less alkyl groups	2
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85. O/N 19/P23/Q1(e)

1(e)(i)	ultra-violet (light) / sunlight	1
1(e)(ii)	$\text{C}_2\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{HCl}$	1

86. O/N 19/P23/Q3(a)(i)

3(a)(i)	cracking	1
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87. O/N 19/P23/Q3(a)(iii)

3(a)(iii)	$\text{C}_4\text{H}_8 + 4\text{O}_2 \rightarrow 4\text{CO} + 4\text{H}_2\text{O}$	1
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88. M/J 19/P21/Q4(c)(ii)

4(c)(ii)	oxidation	1
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89. M/J 19/P22/Q4(b)

4(b)(i)	cracking	1
4(b)(ii)	$2\text{C}_7\text{H}_{16} \rightarrow \text{C}_4\text{H}_{10} + \text{C}_6\text{H}_{14} + 2\text{C}_2\text{H}_4$	1

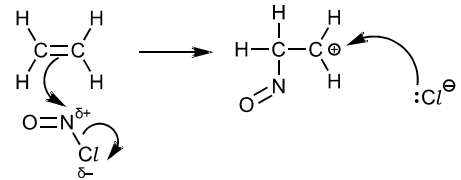
90. M/J 19/P22/Q4(d)(iii)-(iv)

4(d)(iii)	propagation	1
4(d)(iv)	$\text{CH}_4 + (\bullet)\text{Cl} \rightarrow (\bullet)\text{CH}_3 + \text{HCl}$ OR $\text{CH}_4 + (\bullet)\text{Cl} \rightarrow \text{HCl} + \text{CH}_3(\bullet)$	1

91. M/J 19/P23/Q6(a)(i)-(ii)

6(a)(i)	Orange / brown to colourless / decolourises	1
6(a)(ii)	any non-ambiguous structures of:	
	X $(\text{CH}_3)_2\text{C} = \text{CH}_2 / (\text{CH}_3)_2\text{CCH}_2$	1
	Y $\text{CH}_3\text{CH}_2\text{CH} = \text{CH}_2 / \text{C}_2\text{H}_5\text{CHCH}_2$	1
	Z $\text{CH}_3\text{CH} = \text{CHCH}_3 / \text{CH}_3\text{CHCHCH}_3$	1

92. MAR 19/P22/Q1(e)

1(e)	 M1 curly arrow from $\text{C}=\text{C}$ to $\text{N}^{\delta+}$ AND curly arrow from $\text{N}=\text{O}$ to $\text{Cl}^{\delta-}$ M2 intermediate AND curly arrow from lone pair on Cl^- to $\text{C}(+)$	2
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93. MAR 19/P22/Q4(b)(i)-(iii)

4(b)(i)	ultra-violet (light) / sun(light)	1
4(b)(ii)	$\text{CH}_2=\text{CHCH}_3 + \text{Cl}^\bullet \rightarrow \text{CH}_2=\text{CHCH}_2^\bullet + \text{HCl}$ OR $\text{C}_3\text{H}_6 + \text{Cl}^\bullet \rightarrow \text{C}_3\text{H}_5^\bullet + \text{HCl}$	1
4(b)(iii)	free-radical (substitution) reactions are uncontrolled OR further chlorination / substitution occurs	1

94. MAR 19/P22/Q4(c)(i)

4(c)(i)	cold, dilute acidified KMnO_4 / potassium manganate(VII)	1
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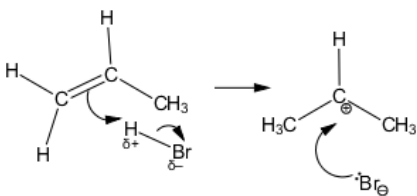
95. MAR 19/P22/Q4(d)

4(d)	M1 major product formed from more stable intermediate / carbocation OR (intermediate has) 2° carbocation which is (more) stable M2 (positive) inductive effect / (+)I of alkyl groups (on the intermediate)	2
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96. O/N 18/P21/Q4(b)(iii)-(iv)

4(b)(iii)	free-radical substitution	1
4(b)(iv)	UV (light) / sunlight	1

97. O/N 18/P22/Q4(d)

4(d)(i)	M1 I experiences a (greater positive) inductive effect due to more alkyl groups OR I contains more electron donating alkyl groups (than II) [1] M2 which stabilises the charge / reduces the charge (on the ion/intermediate) OR spreads the charge across the ion / molecule / intermediate [1]	2
4(d)(ii)	 M1 curly arrow from double bond to H of H—Br [1] M2 curly arrow from H—Br bond to Br AND correct dipole on H—Br [1] M3 curly arrow from lone pair on Br ⁻ to C ⁺ [1]	3
4(d)(iii)	nucleophilic substitution	1

98. O/N 18/P23/Q4(b)(iii)-(iv)

4(b)(iii)	free-radical substitution	1
4(b)(iv)	UV (light) / sunlight	1

99. M/J 18/P21/Q2(b)

2(b)	Produces more useful / more valuable / higher demand substances / alkanes / alkenes	1
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100. M/J 18/P21/Q2(c)(i)

2(c)(i)	$C_{12}H_{26} \rightarrow 2C_2H_4 + C_8H_{18}$	1
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101. M/J 18/P22/Q3(a)

3(a)(i)	combustion	1
3(a)(ii)	$C_8H_{18} + 12\frac{1}{2}O_2 \rightarrow 8CO_2 + 9H_2O$	
	correct species	1
	correct balancing	1

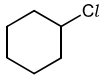
102. MAR 18/P22/Q2(b)(ii)

2(b)(ii)	add bromine (water) / Br ₂ (aq) (brown to) colourless / decolourised	2
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103. MAR 18/P22/Q2(c)(i)

2(c)(i)	addition	1
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104. MAR 18/P22/Q4(a)

4(a)(i)	ultraviolet / UV light	1
4(a)(ii)	initiation HCl propagation 	4

105. MAR 18/P22/Q4(c)(i)

4(c)(i)	acidified AND KMnO ₄ hot AND <u>c</u> (oncentrated)	2
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106. O/N 17/P21/Q3(d)(iii)

3(d)(iii)	X = 2-chlorobutane	1
	Y = 1-chlorobutane	1

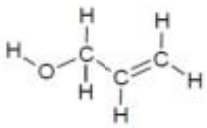
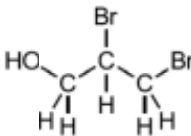
107. O/N 17/P22/Q4(e)

4(e)	2-bromo-2-methylpropane	1
	1-bromo-2-methylpropane	1

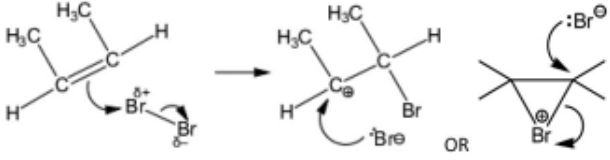
108. O/N 17/P23/Q3(d)(iii)

3(d)(iii)	X = 2-chlorobutane	1
	Y = 1-chlorobutane	1

109. M/J 17/P21/Q4(b)

4(b)(i)		1
4(b)(ii)	(electrophilic) addition	1
	bromine decolourises / turns colourless / fades (from orange / brown)	1
4(b)(iii)	HOCH ₂ CHBrCH ₂ Br OR 	1
4(b)(iv)	CO ₂ / carbon dioxide	1

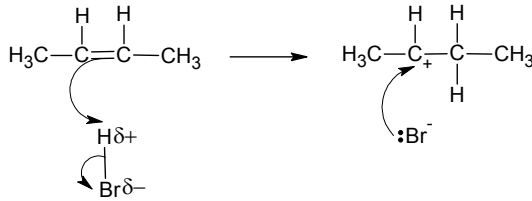
110. M/J 17/P22/Q4(a)(v)-(vi)

4(a)(v)	 arrow from the C=C double bond drawn to the bromine	1
	dipole on Br ₂ in correct orientation AND arrow from the Br-Br bond to the Br ^{δ-}	1
	correct carbocation / bromonium ion from the structure with C=C drawn	1
	Br ⁻ with lone pair, negative charge AND arrow from lone pair to the carbon atom of intermediate OR using both arrows shown (in alternative diagram)	1
4(a)(vi)	electrons in pi bond induce it (the dipole) OR (high) electron density in pi bond / double bond / C=C repels electrons (away from nearest Br) OR polarised by (high) electron density in pi bond / double bond / C=C	1

111. M/J 17/P23/Q4(a)

4(a)	cracking	1
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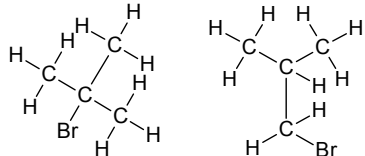
112. M/J 17/P23/Q4(c)(ii)

4(c)(ii)	 arrow from the C=C double bond drawn to the H	1
	dipole on H-Br in correct orientation AND arrow from the H-Br bond to the Br ^{δ-}	1
	correct carbocation from the structure with C=C drawn	1
	Br ⁻ with lone pair, negative charge AND arrow from lone pair to the positively charged carbon atom of intermediate	1

113. M/J 17/P23/Q4(d)(iv)

4(d)(iv)	intermediate / (secondary carbo) cation from X is more stable ora OR charge density of C ⁺ (of the intermediate of X) is reduced	1
	(due to) electron-releasing character / (positive) inductive effect of alkyl groups / / due to electron releasing alkyl group	1

114. M/J 17/P23/Q4(e)(ii)

4(e)(ii)		2
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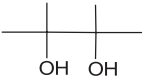
115. MAR 17/P22/Q3(c)(i)

3(c)(i)	(electrophilic) addition	1
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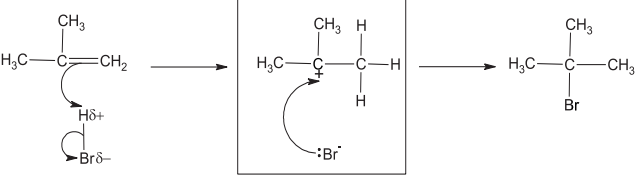
116. MAR 17/P22/Q3(c)(iii)

3(c)(iii)	positive inductive effect of more alkyl groups / more alkyl groups donate electron density	1
	secondary carbocation / secondary intermediate is more stable (than primary)	1

117. O/N 16/P21/Q4(b)(ii)-(iii)

4(b)(ii)		1	1
4(b)(iii)	Propanone	1	1

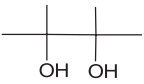
118. O/N 16/P21/Q4(c)(ii)-(iii)

4(c)(ii)		4	4
4(c)(iii)	(tertiary carbocat)ion / (tertiary) intermediate is / C+ with least number of hydrogen atoms bonded to it is more stable (than primary) due to (positive) inductive effect of three/more methyl groups (cf one) / three / more electron releasing methyl groups three / more electron donating methyl groups reducing charge (density) on C+	1 1 1	3

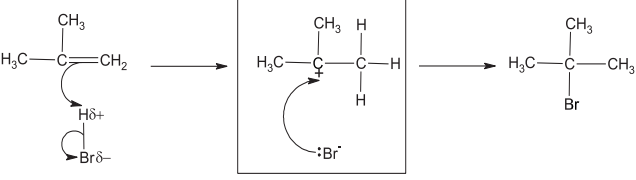
119. O/N 16/P22/Q5(a)(i)

5(a)(i)	$\text{Cl}\cdot$ and $\cdot\text{CH}_3$	1
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120. O/N 16/P23/Q4(b)(ii)-(iii)

4(b)(ii)		1	1
4(b)(iii)	Propanone	1	1

121. O/N 16/P23/Q4(c)(ii)-(iii)

4(c)(ii)		4	4
4(c)(iii)	(tertiary carbocat)ion / (tertiary) intermediate is / C+ with least number of hydrogen atoms bonded to it is more stable (than primary) due to (positive) inductive effect of three/more methyl groups (cf one) / three / more electron releasing methyl groups three / more electron donating methyl groups reducing charge (density) on C+	1 1 1	3

122. M/J 16/P21/Q5(e)(i)

(e) (i)	$V = CH_3CH_2CHCHCH_2CH_3 / CH_3CH_2CH=CHCH_2CH_3$ $T = CH_3CH_2CH(OH)CH(OH)CH_2CH_3$	[1] [1]	[2]
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123. MAR 16/P22/Q4(b)

(b)	$C_8H_{18} + 12\frac{1}{2}O_2 \rightarrow 8CO_2 + 9H_2O$	[1]	[1]
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124. O/N 15/P21/Q3(b)

(b) (i)	$C_7H_{16} + 11O_2 \rightarrow 7CO_2 + 8H_2O$	[1]	[1]
(ii)	$C_7H_{16} + 4O_2 \rightarrow 7C + 8H_2O$	[1]	[1]
(iii)	global dimming / PAN / smog / global warming	[1]	[1]

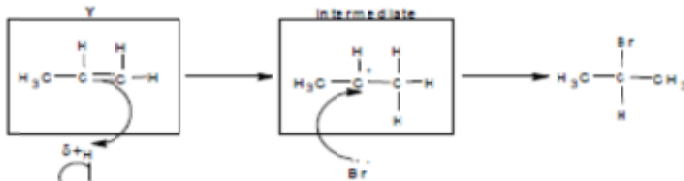
125. O/N 15/P21/Q3(c)

(c) (i)	(Free) Radical Substitution	[1]	[1]
(ii)	$Cl_2 \rightarrow 2Cl\cdot$ OR $Cl_2 \rightarrow Cl\cdot + Cl\cdot$ $C_7H_{16} + Cl\cdot \rightarrow \cdot C_7H_{15} + HCl$ $\cdot C_7H_{15} + Cl_2 \rightarrow C_7H_{15}Cl + Cl\cdot$ $\cdot C_7H_{15} + Cl\cdot \rightarrow C_7H_{15}Cl$ OR $\cdot C_7H_{15} + \cdot C_7H_{15} \rightarrow C_{14}H_{30}$ Initiation; Propagation; Termination (used correctly)	[1] [1] [1] [1]	[5]
			[15]

126. O/N 15/P22/Q4(b)(iv)

(iv)	Reduction LiAlH ₄ / lithium aluminium hydride / lithium tetrahydridoaluminate	(1)	[1] [1]	[2]
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127. O/N 15/P22/Q4(b)(vi)-(viii)

(vi)	 M1 = correct structure of Y and curly arrow from double bond to H M2 = dipole and curly arrow from H-Br bond to Br M3 = correct intermediate M4 = Br⁻ with lone pair and curly arrow from lone pair to C(+)		[1] [1] [1] [1]	[4]
(vii)	electrophilic addition		[1]	[1]
(viii)	secondary carbocation more stable than primary due to electron releasing character / (positive) inductive effect of alkyl groups		[1] [1]	[2]
				[22]

128. O/N 15/P23/Q3(b)

(b) (i)	$C_7H_{16} + 11O_2 \rightarrow 7CO_2 + 8H_2O$	[1]	[1]
(ii)	$C_7H_{16} + 4O_2 \rightarrow 7C + 8H_2O$	[1]	[1]
(iii)	global dimming / PAN / smog / global warming	[1]	[1]

129. O/N 15/P23/Q3(c)

(c) (i)	(Free) Radical Substitution	[1]	[1]
(ii)	$\text{Cl}_2 \rightarrow 2\text{Cl}^\bullet$ OR $\text{Cl}_2 \rightarrow \text{Cl}^\bullet + \text{Cl}^\bullet$ $\text{C}_7\text{H}_{16} + \text{Cl}^\bullet \rightarrow \bullet\text{C}_7\text{H}_{15} + \text{HCl}$ $\bullet\text{C}_7\text{H}_{15} + \text{Cl}_2 \rightarrow \text{C}_7\text{H}_{15}\text{Cl} + \text{Cl}^\bullet$ $\bullet\text{C}_7\text{H}_{15} + \text{Cl}^\bullet \rightarrow \text{C}_7\text{H}_{15}\text{Cl}$ OR $\bullet\text{C}_7\text{H}_{15} + \bullet\text{C}_7\text{H}_{15} \rightarrow \text{C}_{14}\text{H}_{30}$ Initiation; Propagation; Termination (used correctly)	[1] [1] [1] [1] [1]	 [5] [15]

130. M/J 15/P22/Q3(b)(iii)

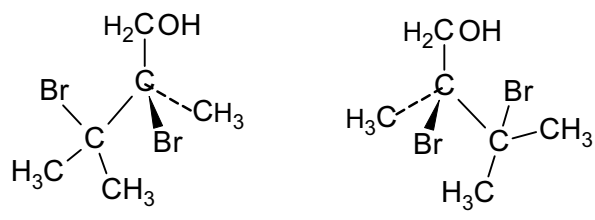
(iii)	H_2 AND Pt or Ni (catalyst)	[1]	[1]
			[13]

131. M/J 15/P23/Q4(a)

4 (a) (i)	$\begin{array}{c} \text{H}_3\text{C} \quad \text{CH}_2\text{OH} \\ \quad \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{CH}_3 \\ \quad \\ \text{HO} \quad \text{OH} \end{array}$	[1]	[1]
(ii)	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C}-\text{C}=\text{O} \end{array}$ $\begin{array}{c} \text{COOH} \\ \\ \text{O}=\text{C} \\ \\ \text{CH}_3 \end{array}$	[1] [1]	 [2]

132. M/J 15/P23/Q4(b)

(b) (i)	M1 = 2 curly arrows M2 = intermediate ion M3 = Br with -ve charge, lone pair and curly arrow to C+	[1] [1] [1]	[3]
(ii)	dipole is <u>induced</u> by proximity to C=C	[1]	[1]
(iii)	Optical	[1]	[1]

(iv)		[1+1]	[2]
			[10]